
PHD FIRST YEAR ANNUAL REVIEW REPORT: RESEARCH ON GAMIFICATION TO ENHANCE USER MOTIVATION IN EXTENDED REALITY (XR) REHABILITATION

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1 Introduction

Extended reality (XR), which includes virtual reality (VR), augmented reality (AR), and mixed reality (MR) [1], has received significant interest in various fields, particularly in healthcare [2]. XR technologies show great promise in transforming rehabilitation processes by providing immersive and interactive environments specifically designed for individuals with physical or cognitive impairments [3–5].

In the field of rehabilitation, XR has been widely used in various areas to cater to a broad spectrum of therapeutic requirements. Applications of XR interventions have been applied in the context of musculoskeletal rehabilitation [6], neurological rehabilitation [3], cognitive rehabilitation [7], and psychological therapy [8]. These interventions have been adopted at addressing conditions such as stroke [9], spinal cord injuries [10], amputations [11], brain injuries [12], and mental health issues [13]. The adaptable nature of XR technology allows for personalized treatments that are specifically designed to meet the unique needs and challenges of individuals in rehabilitation [14].

Despite the potential benefits of XR-assisted rehabilitation, patients often face motivational challenges during their treatment [15]. Motivation is typically defined as what explains why humans start, continue, or stop a certain behavior at a specific moment [16]. Ryan and Deci distinguished motivation between intrinsic and extrinsic motivation [17]. Intrinsic motivation is driven by inherent enjoyment, satisfaction, and interest in an activity, thereby resulting in self-determined behaviour. Conversely, extrinsic motivation is characterized by external incentives or rewards that stimulate behaviour. The issue of low motivation in rehabilitation process can stem from various factors, including repetitive nature of rehabilitation exercises [18], weak beliefs about self-capability [19, 20], lack of goal setting [21, 22], incentives [23], enjoyment of exercise [24]. Besides, a key barrier to patient motivation is their knowledge gap regarding the relevance of personal health and the prescribed exercise regimen [25]. Maintaining motivation for consistent participation in rehabilitation exercises is of significance, which affects adherence to the treatment and, ultimately, the efficacy of the intervention [14].

Therefore, to maximize patient motivation in rehabilitation, this research integrates gamification into the XR rehabilitation process. Gamification is referred to as the selective incorporation of game elements into an interactive system without a fully-fledged game as the end product [26]. In general, anything that is used in the design of a game is often referred to as a game element. However, in most game design frameworks or theories, there are clearer categorizations of these elements. According to Werbach and Hunter, there are three categories of game elements that are relevant to gamification: dynamics, mechanics, and components [27]. Dynamics are the big-picture aspects of the gamified system that ultimately drive the players further. Mechanics are the basic processes that drive the action forward and generate player engagement. Game components are the specific features that represent the intended mechanics and dynamics. Figure 1 demonstrate the Game Element Hierarchy proposed by Werbach and Hunter [27]. Furthermore, the Octalysis Framework created by Chou [28] can be used to investigate how game elements align with eight different core drives for humans motivation. As shown in the figure 2, the framework comprises eight core drives represented by an octagon shape, including Epic Meaning, Accomplishment, Empowerment, Ownership, Social Influence, Scarcity,

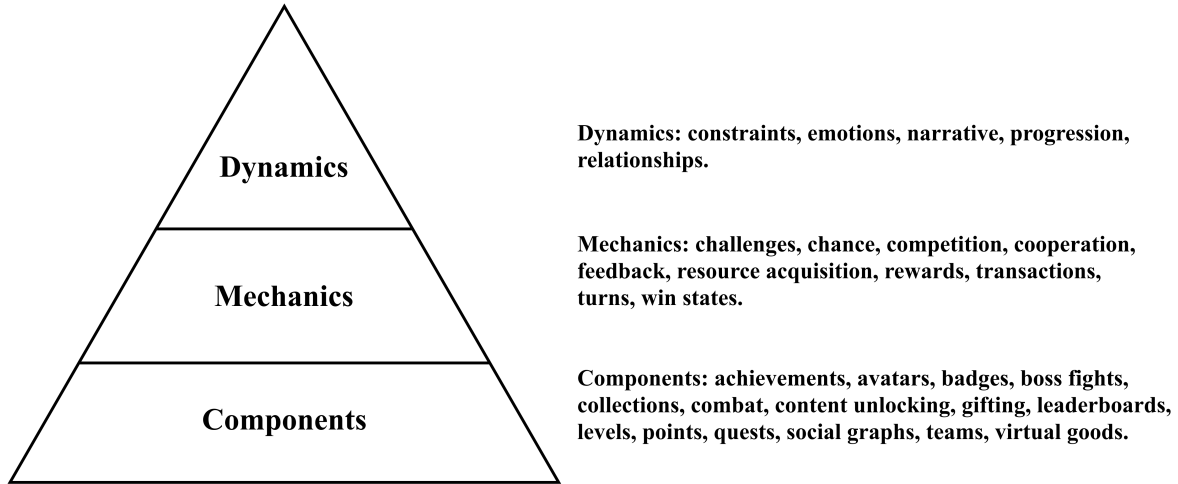


Figure 1: The Game Element Hierarchy [27]

Unpredictability, and Avoidance. By understanding and implementing these core drives in the XR rehabilitation settings can create engaging experiences that cater to motivators and promote positive user experiences.

This PhD research focuses on constructing a user motivation enhancement framework in gamified rehabilitation. The main tasks consist of research on the effect of various game elements on user motivation in MR rehabilitation settings, and how these designed game elements could be improved to further enhance user motivation. The detailed research proposal is presented in Section 4).

The research outcome of my first year PhD study is as follows:

- Han, R., Lee, B.G., Towey, D., Yao, Y., Pike, M. (2024). Exploring Emotional Responses with Dynamic Difficulty Adjustment Adaptation in Immersive Virtual Reality Exergaming. In 2024 IEEE 48th Annual Computers, Software, and Applications Conference (COMPSAC) IEEE. (Accepted)
- Lee, B.G., Wang, X., Han, R., Sun, L., Pike, M., Chung, WY. (2024). Deep Learning Approach for Enhanced Object Recognition and Assembly Guidance with Augmented Reality. In: Choi, B.J., Singh, D., Tiwary, U.S., Chung, WY. (eds) Intelligent Human Computer Interaction. IHCI 2023. Lecture Notes in Computer Science, vol 14532. Springer, Cham. https://doi.org/10.1007/978-3-031-53830-8_11

2 Literature Review

2.1 XR in rehabilitation

Recent developments in XR have shown great potential for making rehabilitation more engaging. Immersive VR (IVR)-based rehabilitation allows patients to carry out rehabilitative exercises while immersing themselves in an interactive virtual environment, increasing the level of engagement which can lead to better therapeutic outcomes [29]. Moan *et al.* [30] conducted a qualitative study to delve into the experiences of stroke survivors and clinicians with a fully IVR treadmill program for stroke rehabilitation, revealing that participants found safety and achievement. Elor *et al.* [31] investigated the effects of a two-month IVR physical therapy program for upper-extremity motions, emphasizing improved user experience and progress evaluation. AR-based rehabilitation introduces a dynamic component to rehabilitation by overlaying digital information onto the physical environment, making therapeutic exercises more personalized. Rodrigues *et al.* [32] developed a prototype of a rehabilitation system that combines a collaborative robot with AR equipment to create a customise rehabilitation experience for patients. Stammeler *et al.* [33] developed an AR-based app (Negami) for the treatment of spatial neglect that combines visual exploration training with rotation of the eyes, head, and trunk. Through patients' natural interaction with the physical surrounding environment during playful tasks, side effects as symptoms of cybersickness are minimised and patients' treatment outcome appeared to markedly increase. MR-based rehabilitation combines real-world and virtual environment in which patients can directly interact with virtual objects. Stamm *et al.* [34] conducted a study on the usability and acceptance among older hypertensive

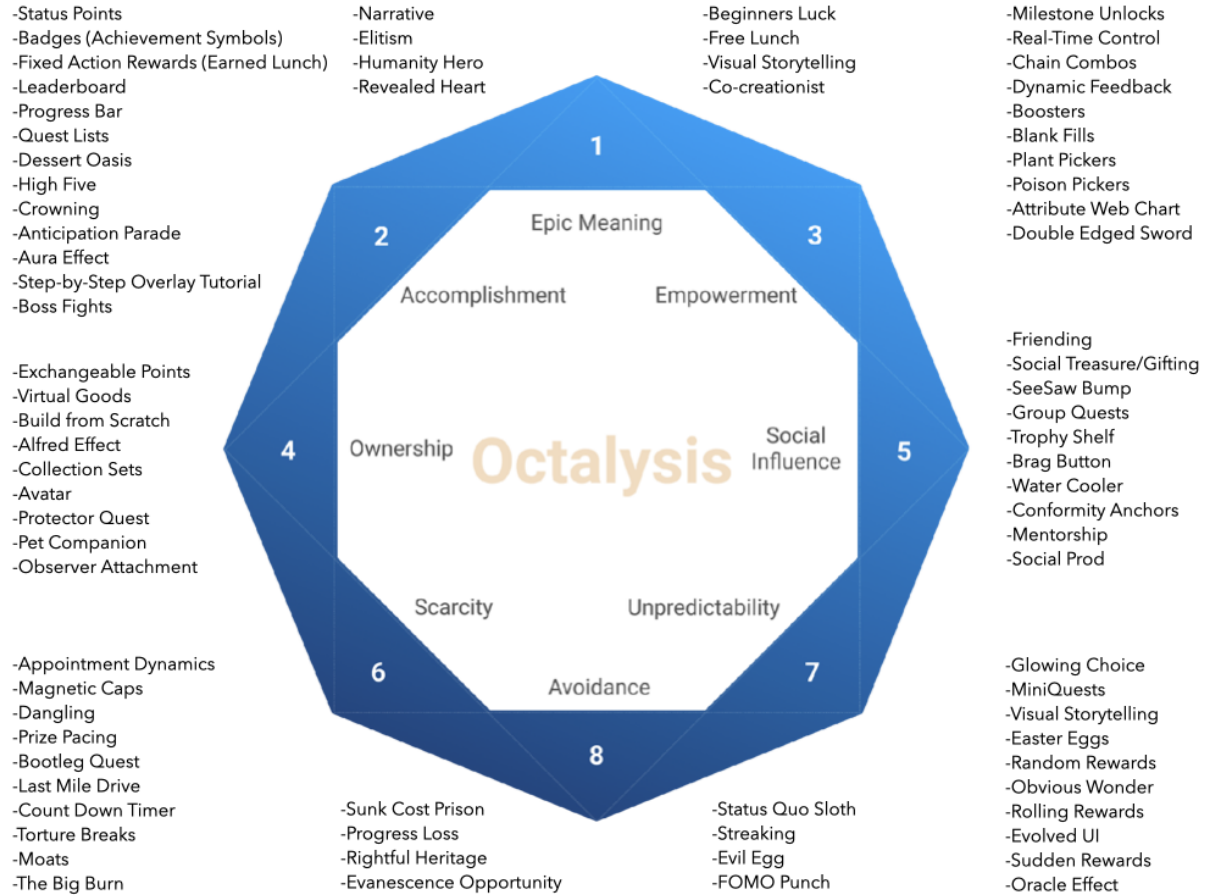


Figure 2: The Octalysis Framework [28]

patients using different types of training in a simulated MR environment. Lancere et al. [11] used MR technologies and wireless sensors to enhance muscle engagement in lower extremity amputee rehabilitation, focusing on the effectiveness of verbal and visual MR feedback in improving muscle activation.

Even though XR provides immersive and engaging environments that can potentially enhance rehabilitation outcomes, patients' motivational challenges and adherence issues remain significant hurdles in rehabilitation that can limit the effectiveness of treatment [15]. Driven by this limitation, this research consider integrating gamification with XR in rehabilitation to enhance patient motivation.

2.2 Gamification

In general, gamification is used to describe those features of an interactive system that aim to motivate and engage end-users through the use of game elements and mechanics [26]. It is a multidisciplinary tool spanning a breadth of domains, theories of thought, methodology, and reasons for implementation. Empirical studies across various disciplines are examining how gamification can be applied in specific contexts and its short- and long-term effects on behavior and experience [26].

Gamification in healthcare encompasses a broad range of applications, from personal care to professional training. Cafazzo *et al.* [35] developed a mobile application to assist children with Type 1 diabetes by utilizing design principles derived from interviews with the children and their caregivers. This application awarded points for recording blood glucose measurements throughout the day. Although rewards were frequent, averaging eight per day, some high achievers chose not to redeem their points, indicating that the rewards did not uniformly motivate all users. Nevertheless, there was a notable 50% increase in daily blood glucose testing. Hori *et al.* [36] developed an application aimed at enhancing social interactions by encouraging users to smile more, which also provided visual and vibrational feedback

for smiles. Two prototypes were developed over a series of studies investigating the effects of the app in different contexts. Results showed that users were motivated to make their friends and themselves smile, but this effect did not translate to the other person or people involved in the social interaction. Even when users' social skills were low, they were perceived by conversational partners to be attentive. Rose *et al.* [37] examined the impact of the mySugr app, a mobile diabetes monitoring tool, on user compliance. They noted that a significant percentage of diabetic individuals either fail to log or inaccurately enter their data. With this app, testing frequency improved by 10-20%, blood sugar levels dropped by 0.4-1.1%, and users reported a higher quality of life. After three months, 85% of the participants were still actively using the app. In an effort to engage young cancer patients in the process of writing a pain diary, Stinson *et al.* [38] developed an app called "Pain Squad" that has users take on the role of recruits on the lookout for pain, represented as a tangible antagonist. Users "rise through the ranks" by capturing their pain records in twice-a-day missions, and obtaining badges and rewards, such as videos by actors from popular cop series, after filling out a certain number of reports. The researchers first designed a cancer-centred pain questionnaire based on an arthritis diary for young people. Two prototypes – one lowfidelity and one high-fidelity – were then developed and tested in a usability study. The researchers then incorporated the usability results into a final prototype for evaluation in a clinical feasibility test. Average compliance was 88%; participants reported day and night, on weekdays or weekends, and consistently over the twoweek period, with no statistical differences; there were no gender differences or differences based on initial treatment location. Satisfaction was high, the app was considered easy to use and did not intrude into daily activities.

Despite the extensive research on the role of gamification in healthcare, there is still a need for further research on the impact of combining XR technologies with gamification, particularly MR.

2.3 Integration of Gamification with XR in rehabilitation

Khaleghi *et al.* [39] explored an IVR-assisted gamified binocular application to treat amblyopia in adults, aiming to improve visual acuity, motivation, and engagement. Izountar *et al.* [14] introduced an IVR adaptive exergame system that considers patient emotions during rehabilitation exercises, with potential benefits for improving rehabilitation outcomes. Chatterjee *et al.* [7] conducted a user study using a custom-built IVR environment called VIRTUE for the cognitive rehabilitation of stroke patients, incorporating gamification techniques to enhance motivation. Knobel *et al.* [40] developed a visual search task embedded in a serious game setting for IVR called the "bird search task". The difficulty level of the game automatically adjusted based on the performance in the previous round, ensuring an optimal task difficulty level for each participant. Shah *et al.* [41] introduced a collaborative exergame in social VR for elderly rehabilitation, emphasising the role of collaboration in motivating exercise and promoting social interaction. Kern *et al.* [42] combined IVR and gamification for gait rehabilitation, in which the custom-designed game elements of cooperation, rewards, collections, and feedback were proven to enhance patient motivation. A clinically-related gesture recognition interactive tool for an AR rehabilitation system proposed by Da Gama *et al.* [43] is proved to be efficient improving the user engagement and exercise performance outcomes. The work of Gmez-Portes *et al.* [44] highlighted how the combination of MR and gamification significantly influences the accuracy of rehabilitation exercises previously defined by therapists. The system developed by Pillai *et al.* [45] successfully provided rehabilitation exercises for upper limb injuries through interactive MR games. The results and comparisons show that this system is relevant, accurate, and superior to previous IVR-based rehabilitation methods because the IVR-based system is blind to the surroundings, whereas the proposed approach has spatial awareness of the environment.

Based on the comprehensive literature review, this PhD research emphasizes MR as the most promising approach for gamified rehabilitation. MR allows for the seamless integration of virtual game elements into the physical rehabilitation environment. This integration can make exercises more relevant and engaging for patients. Furthermore, MR can simulate potentially challenging physical tasks that might be unsafe or impractical to perform in a purely physical setting. This capability is crucial for patients with severe mobility limitations or those in remote locations lacking specialized rehabilitation facilities.

Despite the fact that many studies have employed gamification with XR technology in the field of rehabilitation, the specific game elements that can enhance patient motivation need to be further researched. As this PhD research progresses, a novel framework of user motivation enhancement in gamified MR rehabilitation will be created based on the research of specific game elements.

3 Pilot study

My first year PhD research mainly focuses on the users' emotional responses with DDA adaptation in IVR exergaming, which includes desining an IVR exergaming that integrates facial expression detection, conducting user study, collecting and analyzing results. The detail of this work is shown in Appendix 1: Manuscript "*Exploring Emotional Responses with Dynamic Difficulty Adjustment Adaptation in Immersive Virtual Reality Exergaming*". Furthermore, a review

paper which summarized methods to enhance user motivation in XR rehabilitation is demonstrated in Appendix 2: Manuscript "*Enhancing User Motivation in Extended Reality-Assisted Rehabilitation: A Comprehensive Systematic Review*".

4 A proposal for the research that will be undertaken during the remaining PhD time

4.1 Research objectives

In this PhD research, I consider MR as a complex media platform for delivering rehabilitation programs (consisting of software, hardware, interaction techniques and contents) and consider integrating gamification into the MR rehabilitation programs. The research objectives consist of research on the effect of game elements on user motivation in MR rehabilitation settings, and how these designed game elements could be improved to further enhance user motivation. Following this view, a novel framework of user motivation enhancement in gamified MR rehabilitation will be created.

Following the research objectives, each of the studies in the PhD research address distinct contributions. At the general level, two interrelated research questions need to be answered:

RQ1: What is the effect of various game elements on user motivation in mixed reality rehabilitation settings?

RQ2: How designed game elements can be improved to further enhance user motivation in mixed reality rehabilitation settings?

4.2 Research tasks

To address the two proposed research questions, the following research tasks will be undertaken. The overview of these tasks is illustrated in Figure 3. The process includes game elements design, integration of game elements into MR rehabilitation settings, evaluation of the designed game elements, improvement of the game element design, and evaluation of the improved game elements design. By following this sequence, a novel framework aimed at enhancing user motivation in gamified MR rehabilitation settings will be developed.

4.2.1 Research tasks to address RQ1

Game elements design The design of game elements will draw from established frameworks of Werbach's Game Elements Hierarchy [27] and Chou's Octalysis Framework [28]. These frameworks help categorize game elements effectively. The rehabilitation process in this context involves two primary tasks:

- Design game elements that educate patients on the importance of performing rehabilitation exercises: This process focus more on game elements that derived from extrinsic motivators – patients are motivated because they want to obtain something, whether it be a goal, a good, or anything they cannot obtain.
- Design game elements that maintain and enhance motivation for rehabilitation exercises: This process focus more on game elements that derived from intrinsic motivators - patients do not need a goal or reward to use their creativity, hang out with friends, or feel the suspense of unpredictability, as the activity itself is rewarding on its own.

For the first task, the focus is on game elements derived from extrinsic motivators, where patients are motivated by external rewards such as achieving goals. The second task focuses on game elements derived from intrinsic motivators, in which the activity itself is rewarding. This includes elements that foster creativity, social interaction, or the thrill of unpredictability, such as engaging storylines or creative tasks that patients find enjoyable and fulfilling on their own.

Based on the Octalysis Framework, the rehabilitation process involves designing game elements for four phases: Discovery, Onboarding, Scaffolding, and Endgame. In the Discovery phase, the goal is to attract users to the rehabilitation program using elements that invoke curiosity, such as engaging introductory stories or initial challenges that highlight the program's benefits. The Discovery phase matches with the first primary task, in which patients are educated on the importance of rehabilitation. In the Onboarding phase, the focus shifts to helping users understand and feel comfortable with the system, providing clear instructions, tutorials, and easy initial tasks to build confidence. During the Scaffolding phase, the aim is to keep users engaged by introducing increasingly difficult tasks, diverse activities, and social elements, etc. This might involve regularly updated exercise routines with varying difficulty levels and social leaderboards to encourage patients. Finally, in the Endgame phase, the objective is to sustain long-term engagement by offering advanced challenges, opportunities for mastery, and recognition for long-term participation, such as advanced levels that unlock after mastering basic exercises, coupled with badges for long-term commitment. The Onboarding,

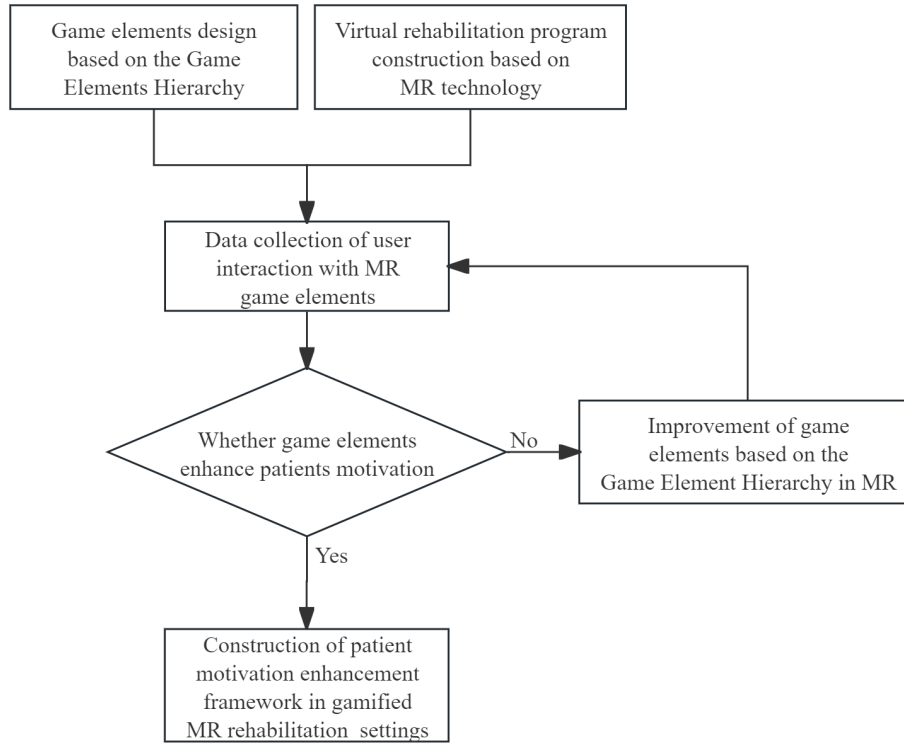


Figure 3: Overview of research tasks

Scaffolding, and Endgame phases match with the second primary task, in which patients are motivated by the game elements during rehabilitation exercises.

To further personalize the rehabilitation experience, the Hexad User Types model [46] categorizes users into six types based on their motivational drivers: Achievers, Explorers, Socializers, Philanthropists, Players, and Disruptors. For Achievers, who are motivated by mastery and achieving goals, the program might include badges, levels, and achievements for completing exercises and reaching milestones. Explorers, motivated by discovery, could engage with hidden features and new environments. Socializers, who thrive on social interaction, might benefit from group challenges and collaboration with others. Philanthropists, motivated by purpose, could have opportunities to mentor new patients or share success stories. Players, driven by rewards, could earn points or virtual currency for completing tasks. Disruptors, who seek change and innovation, might enjoy customizing the virtual environment or proposing new exercises. By designing different game elements for these 6 groups of patients can make the rehabilitation process more engaging and personalized.

Integration of game elements into MR rehabilitation settings In this research, MR is considered as rehabilitation settings. MR allows for the seamless incorporation of virtual elements into real-world settings, enabling simulations of complex and potentially unsafe physical tasks in a safe and controlled environment. This research will focus on how the peculiarities of MR can be combined with rehabilitation game elements.

Evaluation of the designed game elements The evaluation of the designed game elements will employ both qualitative and quantitative methods. This evaluation aims to understand the impact of various game elements on user motivation within MR settings. Specifically, it includes:

- Evaluation of users' subjective responses: this will use User Experience Questionnaire (UEQ) [47], Intrinsic Motivation Inventory (IMI) [48], NASA Task Load Index (NASA-TLX) [49] to measure their experience and motivation of the designed game elements.

- Evaluation of users' physiological responses: this includes EEG monitoring, which captures brain activity, offering insights into cognitive load and emotional states.

Based on the Octalysis Framework, the evaluation will focus on intrinsic and extrinsic motivation core drives. The initial evaluation will lead to the development of a preliminary user motivation framework, which identifies both effective and ineffective game elements.

4.2.2 Research tasks to address RQ2

Improvement of game elements design Based on the findings from RQ1, the next phase will involve refining and improving game elements to enhance user motivation further. This task will focus on analyzing elements that had a weak or negative impact on motivation to understand why they failed, and then using insights from the initial evaluation to redesign these elements to better align with the motivational needs of users. Incorporating direct feedback from users will be crucial to inform these design improvements. This iterative design process aims to enhance the overall motivational framework by continuously refining game elements based on empirical evidence.

Evaluation of the improved game elements design The improved game elements will undergo a second round of evaluation to compare their effectiveness against the initial designs. This comparative analysis will involve using the same evaluation methods to measure changes in user motivation, gathering detailed feedback through surveys, interviews, and focus groups to understand the subjective improvements in user experience. This phase will generate comprehensive results of the involved game elements.

4.2.3 A novel framework for enhancement of user motivation

The final task will synthesize the findings from RQ1 and RQ2 to develop a novel framework for enhancing user motivation in gamified MR rehabilitation settings. This framework will integrate game elements proven to enhance motivation and exclude or significantly improve elements that had a negative impact on motivation. It will also provide clear guidelines for designing gamified MR rehabilitation programs based on empirical evidence. The final framework will be a comprehensive guide for future development in the field, which aims at maximizing user motivation through well-designed game elements.

4.3 A time plan

A Gantt chart for the time plan of my remaining works is shown in Figure 4.

4.3.1 Year 2024 - 2025

Year 2024 Q2 - Q3: Initial game elements design

- Conduct a comprehensive literature review on Werbach's Game Element Hierarchy and Chou's Octalysis Framework, as well as their related applications.
- Initially identify key game elements in MR settings.
- Design game elements that educate patients on the importance of performing rehabilitation.
- Design game elements that maintain and enhance motivation for rehabilitation exercises.
- Evaluation of the initial game elements design.
- Plan to submit a paper to CHI conference.

Year 2024 Q4 - Year 2025 Q2: Game elements design for different user types

- Design game elements that tailored to 6 different user types based on the Hexad User Types model.
- Comprehensive evaluation of the game elements design for 6 different user types.
- Plan to submit a paper to IJHCS journal.

4.3.2 Year 2025 - 2026

Year 2025 Q3 - Q4: Improvement of game elements design

- Based on the findings from RQ1, refine and improve game elements to enhance user motivation further.

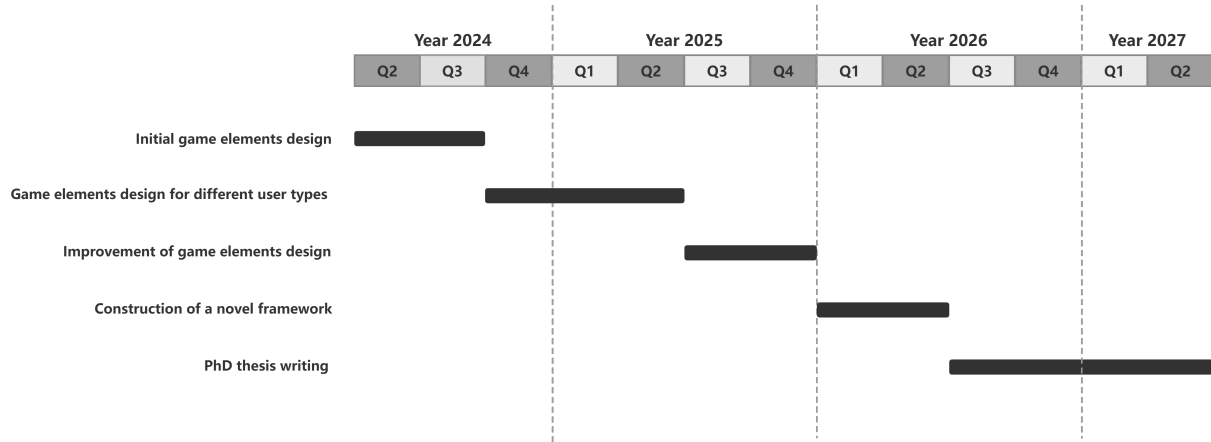


Figure 4: Time Plan Gantt Chart

- Evaluation of the effectiveness of improved game elements against the initial designs.
- Plan to submit a paper to TOCHI journal.

Year 2026 Q1 - Q2: Construction of a novel framework for user motivation enhancement

- Based on findings from RQ1 and RQ2, develop a novel framework for enhancing user motivation in gamified MR rehabilitation settings.
- Evaluation of the proposed framework.
- Plan to submit a paper to IJHCS journal.

4.3.3 Year 2026 - 2027

Year 2026 Q3 - Year 2027 Q2: PhD thesis writing

- Synthesis the whole PhD study and write a complete thesis for this research.

5 Conclusion

This PhD research investigates the integration of gamification with MR to enhance user motivation in rehabilitation settings. The study addresses two primary questions: the impact of various game elements on user motivation and the optimal design of these elements to further enhance motivation. Through a structured and iterative process, the research will design, integrate, evaluate, and improve game elements, ultimately developing a novel framework that combines MR and gamification to make rehabilitation more engaging and effective. This framework aims to provide valuable insights and guidelines for future development in gamified MR rehabilitation, with the goal of maximizing patient motivation and improving therapeutic outcomes.

References

- [1] Christopher Andrews, Michael K Southworth, Jennifer NA Silva, and Jonathan R Silva. Extended reality in medical practice. *Current treatment options in cardiovascular medicine*, 21:1–12, 2019.
- [2] Anitha S Pillai and Prabha Susy Mathew. Impact of virtual reality in healthcare: a review. *Virtual and augmented reality in mental health treatment*, pages 17–31, 2019.
- [3] Rocco Salvatore Calabrò, Antonio Cerasa, Irene Ciancarelli, Loris Pignolo, Paolo Tonin, Marco Iosa, and Giovanni Morone. The arrival of the metaverse in neurorehabilitation: Fact, fake or vision? *Biomedicine*, 10(10):2602, 2022.

- [4] Ilona M Punt, Stéphane Armand, Jean-Luc Ziltener, and Lara Allet. Effect of wii fit™ exercise therapy on gait parameters in ankle sprain patients: A randomized controlled trial. *Gait & posture*, 58:52–58, 2017.
- [5] CA DiCesare, AW Kiefer, SH Bonnette, and GD Myer. Realistic soccer-specific virtual environment exposes high-risk lower extremity biomechanics. *J Sport Rehabil*, 29(3):294–300, 2019.
- [6] Joke Schuermans, Alena Van Hootegem, Margo Van den Bossche, Marjolein Van Gendt, Erik Witvrouw, and Evi Wezenbeek. Extended reality in musculoskeletal rehabilitation and injury prevention-a systematic review. *Physical therapy in sport*, 55:229–240, 2022.
- [7] Kausik Chatterjee, Alastair Buchanan, Katy Cottrell, Sara Hughes, Thomas W Day, and Nigel W John. Immersive virtual reality for the cognitive rehabilitation of stroke survivors. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 30:719–728, 2022.
- [8] Carola Gatto, Giovanni D’Errico, Fabiana Nuccetelli, Valerio De Luca, Giovanna Ilenia Paladini, and Lucio Tommaso De Paolis. Xr-based mindfulness and art therapy: facing the psychological impact of covid-19 emergency. In *Augmented Reality, Virtual Reality, and Computer Graphics: 7th International Conference, AVR 2020, Lecce, Italy, September 7–10, 2020, Proceedings, Part II* 7, pages 147–155. Springer, 2020.
- [9] Ying Dong, Xiaoyu Liu, Min Tang, Hongqiang Huo, Duo Chen, Zhixin Wu, Ran An, and Yubo Fan. A haptic-feedback virtual reality system to improve the box and block test (bbt) for upper extremity motor function assessment. *Virtual Reality*, 27(2):1199–1219, 2023.
- [10] Marco Solcà, Vibhor Krishna, Nicole Young, Milind Deogaonkar, Bruno Herbelin, Pavo Orepic, Robin Mange, Giulio Rognini, Andrea Serino, Ali Rezai, et al. Enhancing analgesic spinal cord stimulation for chronic pain with personalized immersive virtual reality. *Pain*, 162(6):1641–1649, 2021.
- [11] L Lancere, M Jürgen, and H Gapeyeva. Mixed reality and sensor real-time feedback to increase muscle engagement during deep core exercising. *Virtual Reality*, pages 1–15, 2023.
- [12] Stephanie MN Glegg, Liisa Holsti, Sue Stanton, Steven Hanna, Diana Velikonja, Barbara Ansley, Denise Sartor, and Christine Brum. Evaluating change in virtual reality adoption for brain injury rehabilitation following knowledge translation. *Disability and Rehabilitation: Assistive Technology*, 12(3):217–226, 2017.
- [13] Patricia Pons, Samuel Navas-Medrano, and Jose L Soler-Dominguez. Extended reality for mental health: Current trends and future challenges. *Frontiers in Computer Science*, 4:1034307, 2022.
- [14] Yousra Izountar, Samir Benbelkacem, Samir Otmane, Abdallah Khababa, Mostefa Masmoudi, and Nadia Zenati. Vr-peer: A personalized exer-game platform based on emotion recognition. *Electronics*, 11(3):455, 2022.
- [15] Daekook M Nekar, Hye Yun Kang, and Jae Ho Yu. Improvements of physical activity performance and motivation in adult men through augmented reality approach: a randomized controlled trial. *Journal of Environmental and Public Health*, 2022, 2022.
- [16] Brenda Major, Catherine Cozzarelli, Mardi J Horowitz, Peter J Colyer, Lynn S Fuchs, Edward S Shapiro, Karen Callan Stoiber, Ulrik Fredrik Malt, Thomas Teo, David G Winter, et al. Encyclopedia of psychology: 8 volume set. *New York and Washington: Oxford University Press and the American Psychological Association*, 2000.
- [17] Richard M Ryan and Edward L Deci. Intrinsic and extrinsic motivation from a self-determination theory perspective: Definitions, theory, practices, and future directions. *Contemporary educational psychology*, 61:101860, 2020.
- [18] Ya-Xuan Hung, Pei-Chen Huang, Kuan-Ta Chen, and Woei-Chyn Chu. What do stroke patients look for in game-based rehabilitation: a survey study. *Medicine*, 95(11):e3032, 2016.
- [19] SM Nyenhuis, N Shah, J Ma, DX Marquez, J Wilbur, A Cattamanchi, and LK Sharp. Identifying barriers to physical activity among african american women with asthma. *Cogent medicine*, 6(1):1582399, 2019.
- [20] Sharmilee M Nyenhuis, Guilherme Moraes Balbim, Jun Ma, David X Marquez, JoEllen Wilbur, Lisa K Sharp, and Spyros Kitsiou. A walking intervention supplemented with mobile health technology in low-active urban african american women with asthma: proof-of-concept study. *JMIR formative research*, 4(3):e13900, 2020.
- [21] Deirdre MJ Walsh, Kieran Moran, Véronique Cornelissen, Roselien Buys, Nils Cornelis, and Catherine Woods. Electronic health physical activity behavior change intervention to self-manage cardiovascular disease: qualitative exploration of patient and health professional requirements. *Journal of medical Internet research*, 20(5):e163, 2018.
- [22] Karin Riemann-Lorenz, Robert W Motl, Bláthin Casey, Susan Coote, Anne Daubmann, and Christoph Heesen. Possible determinants of long-term adherence to physical activity in multiple sclerosis—theory-based development of a comprehensive questionnaire and results from a german survey study. *Disability and rehabilitation*, 43(22):3175–3188, 2021.

- [23] Rebecca L Morris, Keith D Hill, Ilana N Ackerman, Darshini Ayton, Glenn Arendts, Caroline Brand, Peter Cameron, Christopher D Etherton-Beer, Leon Flicker, Anne-Marie Hill, et al. A mixed methods process evaluation of a person-centred falls prevention program. *BMC health services research*, 19:1–15, 2019.
- [24] Pia Andersen, Lena Lendahls, Sara Holmberg, and Per Nilsen. Patients’ experiences of physical activity on prescription with access to counsellors in routine care: a qualitative study in sweden. *BMC public health*, 19:1–11, 2019.
- [25] Junsheng L Teo, Zhen Zheng, and Stephen R Bird. Identifying the factors affecting ‘patient engagement’ in exercise rehabilitation. *BMC Sports Science, Medicine and Rehabilitation*, 14(1):18, 2022.
- [26] Katie Seaborn and Deborah I Fels. Gamification in theory and action: A survey. *International Journal of human-computer studies*, 74:14–31, 2015.
- [27] Kevin Werbach, Dan Hunter, and Walter Dixon. *For the win: How game thinking can revolutionize your business*, volume 1. Wharton digital press Philadelphia, 2012.
- [28] Yu-kai Chou. *Actionable gamification: Beyond points, badges, and leaderboards*. Packt Publishing Ltd, 2019.
- [29] Desiderio Cano Porras, Petra Siemonsma, Rivka Inzelberg, Gabriel Zeilig, and Meir Plotnik. Advantages of virtual reality in the rehabilitation of balance and gait: systematic review. *Neurology*, 90(22):1017–1025, 2018.
- [30] Merete Endresen Moan, Elise Klæbo Vonstad, Xiaomeng Su, Beatrix Vereijken, Marit Solbjør, and Nina Skjæret-Maroni. Experiences of stroke survivors and clinicians with a fully immersive virtual reality treadmill exergame for stroke rehabilitation: a qualitative pilot study. *Frontiers in Aging Neuroscience*, 13:735251, 2021.
- [31] Aviv Elor, Michael Powell, Evanjin Mahmoodi, Mircea Teodorescu, and Sri Kurniawan. Gaming beyond the novelty effect of immersive virtual reality for physical rehabilitation. *IEEE Transactions on Games*, 14(1):107–115, 2021.
- [32] José Carlos Rodrigues, Paulo Menezes, and Maria Teresa Restivo. An augmented reality interface to control a collaborative robot in rehab: A preliminary usability evaluation. *Frontiers in Digital Health*, 5:1078511, 2023.
- [33] Britta Stammer, Kathrin Flammer, Thomas Schuster, Marian Lambert, and Hans-Otto Karnath. Negami: An augmented reality app for the treatment of spatial neglect after stroke. *JMIR Serious Games*, 11:e40651, 2023.
- [34] Oskar Stamm, Susan Vorwerk, Michele Haink, Kristian Hildebrand, and Ilona Buchem. Usability and acceptance of exergames using different types of training among older hypertensive patients in a simulated mixed reality. *Applied Sciences*, 12(22):11424, 2022.
- [35] Joseph A Cafazzo, Mark Casselman, Nathaniel Hamming, Debra K Katzman, and Mark R Palmert. Design of an mhealth app for the self-management of adolescent type 1 diabetes: a pilot study. *Journal of medical Internet research*, 14(3):e2058, 2012.
- [36] Yukari Hori, Yutaka Tokuda, Takahiro Miura, Atsushi Hiyama, and Michitaka Hirose. Communication pedometer: a discussion of gamified communication focused on frequency of smiles. In *Proceedings of the 4th augmented human international conference*, pages 206–212, 2013.
- [37] KJ Rose, Martin Koenig, and Frank Wiesbauer. Evaluating success for behavioral change in diabetes via mhealth and gamification: Mysugr’s keys to retention and patient engagement. *Diabetes Technology & Therapeutics*, 15(S1):A114, 2013.
- [38] Jennifer N Stinson, Lindsay A Jibb, Cynthia Nguyen, Paul C Nathan, Anne Marie Maloney, L Lee Dupuis, J Ted Gerstle, Benjamin Alman, Sevan Hopyan, Caron Strahlendorf, et al. Development and testing of a multidimensional iphone pain assessment application for adolescents with cancer. *Journal of medical Internet research*, 15(3):e2350, 2013.
- [39] Ali Khaleghi, Zahra Aghaei, and Fateme Hosseinnia. Toward using effective elements in adults’ amblyopia treatment in a virtual reality-based gamified binocular application. *Entertainment Computing*, 43:100504, 2022.
- [40] Samuel Elia Johannes Knobel, Brigitte Charlotte Kaufmann, Stephan Moreno Gerber, Prabitha Urwyler, Dario Cazzoli, René M Müri, Tobias Nef, and Thomas Nyffeler. Development of a search task using immersive virtual reality: proof-of-concept study. *JMIR Serious Games*, 9(3):e29182, 2021.
- [41] Syed Hammad Hussain Shah, Anniken Susanne T Karlsen, Mads Solberg, and Ibrahim A Hameed. A social vr-based collaborative exergame for rehabilitation: codesign, development and user study. *Virtual Reality*, pages 1–18, 2022.
- [42] Florian Kern, Carla Winter, Dominik Gall, Ivo Käthner, Paul Pauli, and Marc Erich Latoschik. Immersive virtual reality and gamification within procedurally generated environments to increase motivation during gait rehabilitation. In *2019 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pages 500–509. IEEE, 2019.

- [43] Alana Elza Fontes Da Gama, Thiago Menezes Chaves, Lucas Silva Figueiredo, Adriana Baltar, Ma Meng, Nassir Navab, Veronica Teichrieb, and Pascal Fallavollita. Mirrabilitation: A clinically-related gesture recognition interactive tool for an ar rehabilitation system. *Computer methods and programs in biomedicine*, 135:105–114, 2016.
- [44] Cristian Gmez-Portes, David Carneros-Prado, Javier Albusac, Jose J Castro-Schez, Carlos Glez-Morcillo, and David Vallejo. Phyre up! a system based on mixed reality and gamification to provide home rehabilitation for stroke patients. *IEEE Access*, 9:139122–139137, 2021.
- [45] Aditya Pillai, Md Samiul Haque Sunny, Md Tanzil Shahria, Nayan Banik, and Mohammad Habibur Rahman. Gamification of upper limb rehabilitation in mixed-reality environment. *Applied Sciences*, 12(23):12260, 2022.
- [46] Gustavo F Tondello, Rina R Wehbe, Lisa Diamond, Marc Busch, Andrzej Marczewski, and Lennart E Nacke. The gamification user types hexad scale. In *Proceedings of the 2016 annual symposium on computer-human interaction in play*, pages 229–243, 2016.
- [47] Martin Schrepp, Andreas Hinderks, and Jörg Thomaschewski. Applying the user experience questionnaire (ueq) in different evaluation scenarios. In *Design, User Experience, and Usability. Theories, Methods, and Tools for Designing the User Experience: Third International Conference, DUXU 2014, Held as Part of HCI International 2014, Heraklion, Crete, Greece, June 22-27, 2014, Proceedings, Part I 3*, pages 383–392. Springer, 2014.
- [48] Richard Ryan and Edward Deci. Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *The American psychologist*, 55:68–78, 01 2000.
- [49] Sandra G Hart. Nasa-task load index (nasa-tlx); 20 years later. In *Proceedings of the human factors and ergonomics society annual meeting*, volume 50, pages 904–908. Sage publications Sage CA: Los Angeles, CA, 2006.

Appendices

Appendix 1: Manuscript "***Exploring Emotional Responses with Dynamic Difficulty Adjustment Adaptation in Immersive Virtual Reality Exergaming***"; Accepted.

Appendix 2: Manuscript "***Enhancing User Motivation in Extended Reality-Assisted Rehabilitation: A Comprehensive Systematic Review***"; Submitted in OCT 2023, receive notification of revise and resubmit in APR 2024, revising now.

Exploring Emotional Responses with Dynamic Difficulty Adjustment Adaptation in Immersive Virtual Reality Exergaming

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Abstract—Immersive Virtual Reality (IVR) exergaming presents a promising avenue to integrate physical exercise with engaging virtual experiences, potentially encouraging sustained physical activity. However, maintaining user motivation over extended periods poses a significant challenge. Recent research has introduced the Dynamic Difficulty Adjustment (DDA) mechanism, dynamically regulating exergame difficulty based on specific conditions to enhance user adaptation. While prior studies have predominantly focused on gaming performance to adjust difficulty, they often overlook the emotional impact on user motivation. This study investigates users' emotional responses to the game timer change (TC) as a DDA mechanism during IVR exergaming. Results indicate that subjects in the TC-implemented game displayed more neutral emotions, concomitant with improved gaming performance. Conversely, subjects in the game without TC exhibited a broader range of detected emotions (sad and happy), suggesting difficulties in adapting to in-game difficulty levels incongruent with their gaming abilities. Overall, this study establishes a foundation for future research in affective computing-based IVR exergaming, aiming to develop an intelligent autonomous DDA mechanism tailored to users' physical and mental conditions.

Index Terms—Immersive virtual reality, dynamic difficulty adjustment, affective computing

I. INTRODUCTION

Exergaming refers to digital games that necessitate physical movements for gameplay, creating an active gaming experience serving as a form of physical activity [1]. It has been shown to positively influence and motivate individuals to participate in physical activity [2]. With the rapid advancement of immersive virtual reality (IVR) technology, exergaming has

evolved from a basic fitness activity to a novel interactive experience [3]. IVR exergaming seamlessly integrates physical exercise with gameplay within an immersive virtual setting, effectively promoting physical activity among users [4], [5]. This approach proved practical for addressing sedentary issues in individuals [6]. Recent research suggested that the immersive nature of IVR exergaming significantly boosted the user's sense of energy and motivation during training, potentially leading to improved aerobic exercise performance [7], [8]. However, an ongoing challenge lies in enhancing user motivation to engage in IVR exergaming. This proves a challenging issue, as several factors including: user fitness levels; preferences; and objectives; must be considered. Consequently, designing adaptable game mechanics that cater to diverse users and sustain their engagement emerged as a pragmatic solution to address this issue.

Hence, a dynamic difficulty adjustment (DDA) mechanism was proposed by [9], which automatically adjusts the game's difficulty level by analyzing the user's performance. This approach proved effective in reducing difficulty levels for beginners, thereby preventing early frustration [10]. Conversely, for more gaming-experienced users, the difficulty level can be elevated to sustain a challenging experience [10]. DDA had been widely implemented in non-immersive VR (NIVR) exergaming to customize game parameters, such as in-game object speed, to accommodate the varying capabilities of individuals in rehabilitation programs [11], [12]. Its integration into IVR exergaming further enhances the personalized and engaging nature of the gaming experience [13]–[15]. Despite its widespread use, there was a paucity of research addressing the emotional behavioral changes of users upon the activation of DDA mechanism in IVR exergaming.

To address this challenge, this work proposes a study on

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the emotional changes of users in response to the dynamically adjustable timer change (TC) feature, a prevalent DDA mechanism, within an IVR exergame. The contributions of this research are outlined as follows:

- Analyzed the emotional changes by examining facial expressions and physiological signals of users during gameplay. The analysis was validated through the evaluation of characteristics from the Intrinsic Motivation Inventory (IMI) [16] among users.
- Explored the correlation between user emotions across two IVR exergaming experiences: one with the TC feature implemented and the other without it.

II. RELATED WORK

A. Exergaming in XR

The promotion of physical activity through exergaming has primarily aimed at addressing sedentary behavior and combating obesity. Engagement in exergaming resulted in a significant increase in heart rate (HR) and energy expenditure compared to sedentary behavior, with little difference observed in comparison to traditional moderate-intensity physical activities [17]. Sween *et al.* [18] identified a robust correlation between exergaming and heightened physical activity levels, predominantly achieving moderate-intensity activity levels in alignment with the American College of Sports Medicine guidelines for health and fitness. Additionally, evidence suggested that longer durations of exergaming may be more beneficial in meeting these recommendations [18].

In a study investigating the impact of IVR exergaming on high-intensity interval training (HIIT), Farrow *et al.* [19] revealed that IVR exergaming significantly enhanced the enjoyment of HIIT sessions for untrained individuals. The research also emphasized that allowing participants to compete against their own previous performance effectively increased exercise intensity in VR-HIIT sessions. Additionally, increasing the resistance in the game, such as the “hard mode”, the highest difficulty level for the game, led to improved cycling performance during HIIT, without compromising participants’ enjoyment. These findings were supported by post-exercise questionnaires assessing intrinsic motivation, subjective vitality, and future exercise intentions. On the other hand, Huang *et al.* [20] explored the impact of IVR exergaming on the executive functions of older adults. The results indicated that participating in an IVR environment positively influenced specific cognitive tasks, with the sense of presence playing a mediating role in this enhancement. These findings underscore the utility and impact of IVR exergaming in enhancing cognitive performance and potentially preventing cognitive decline among users.

B. DDA in XR-based Exergaming

Two fundamental components of the DDA mechanism in XR-based exergaming include the activation conditions of DDA within the exergames and the implementation methods of the DDA adaptation mechanism in the exergames [21]. Typically, the activation conditions of DDA in exergaming

can be categorized into two approaches: performance-based adaptation and physiological-based adaptation.

Performance-based adaptation involves assessing users’ performance during gameplay to adjust the difficulty level of the exergame. This assessment may include factors such as the damage inflicted on virtual enemies or targeted objects [22], the areas and distances covered during the collection of virtual items [23], health point status [24], movement speed and accuracy [25], or accurate responses to fulfill mission tasks [26]. Physiological-based adaptation employs physiological indicators to trigger the DDA mechanism, analyzing factors such as brain activities [27], eye movements [28], changes in HR [29], or facial expressions [30] of users during gameplay. In many instances, these approaches are used to infer mental states, such as excitement, anxiety, or attention levels.

For instance, Stein *et al.* [31] contributed to DDA by developing a method that adjusts difficulty levels based on the excitement level computed from measured EEG signals. The study observed an increase in user excitement in the exergaming with the implemented DDA mechanism compared to exergaming without DDA. Similarly, Fernandez *et al.* [32] integrated the DDA mechanism with Rhythm-Group Theory in a 2D platform exergame, where the exergame difficulty level was adjusted in real-time based on the user’s attention level. Chanel *et al.* [33] proposed the use of deep data-driven affective models capable of detecting anxiety and boredom from the user’s physiological signals to adjust the gaming difficulty and increase user engagement.

The adaptation of the game timer represents a prevalent and common implemented mechanism within DDA. Adjusting the timer’s speed and duration during exergame gameplay holds substantial potential to influence the user’s overall experience [34]. Typically, Galy *et al.* [35] observed that introducing time pressure and modifying task difficulty levels were an effective strategy for enhancing user motivation. Another principal adaptation strategy involves adjusting the in-game virtual object’s speed, although this proves more effective in 2D games compared to more intricate 3D games, possibly due to the simpler design nature of 2D games [28], [33].

Procedural Content Generation (PCG) serves as an alternative method, wherein the level design is procedurally adjusted to either increase or decrease the game difficulty level. Huber *et al.* [36] implemented PCG for DDA in IVR exergaming, dynamically generating game levels tailored to users’ performances and abilities. This approach aims to alleviate monotony over extended periods and enable concurrent adjustments of both cognitive and physical demands during IVR exergaming. Furthermore, research highlights the positive effects of other DDA mechanisms, such as the implementation of varying task complexity [37], in-game feedback [38], and hint-supporting mechanisms [27].

Despite the promising outcomes of DDA in motivating users, there exists a gap in comprehensive research analyzing changes in users’ emotional states concerning the integration of DDA in IVR exergaming. This gap is particularly notable concerning the suitability and comfort of the DDA mecha-

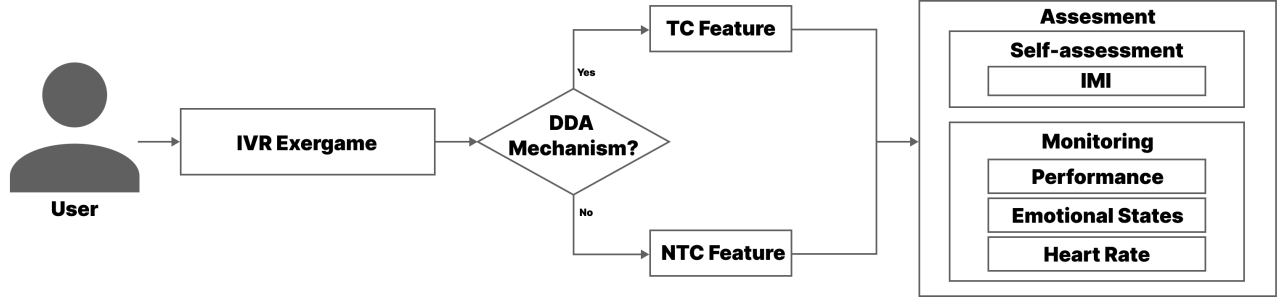


Fig. 1: System overview.



(a)



(b)

Fig. 2: The proposed design of the (a) IVR soccer exergaming and (b) illustration of a participant engaged in the IVR exergaming, utilizing virtual hands through the Meta Quest Pro device.

nism for exergaming users, with a focus on automating the adjustment of game difficulty levels based on users' physical performance, such as their gaming score.

III. METHODS

Figure 1 illustrates the system overview of the IVR exergaming design, including the proposed TC and NTC feature-based DDA mechanism, followed by an assessment of the emotional impact on DDA in IVR exergaming.

A. Design of IVR Exergaming

To conduct the study, a self-designed IVR soccer exergame was developed utilizing Unity3D [39] and integrating the Meta Movement SDK [40], which enables facial feature extraction during gameplay. In this game, the user assumes the role of a goalkeeper positioned in front of a virtual soccer goal. The objective is to prevent soccer balls from entering the goal by using virtual hands to block them as illustrated in Fig. 2a. Each successful defense resulted in the user earning one point.

In this study, the TC feature was used as a DDA mechanism where prior research has underscored TC's efficacy in en-

hancing user motivation throughout exergaming activities [21], [34]. The design and integration of TC within this research drew upon the framework established by [41], with minor refinements tailored to the specifics of this inquiry. The TC adaptation is activated every 10 seconds during gameplay. The TC feature underwent manipulation through a time speed multiplier M , determined based on the gaming score GS and the real-time standard gaming score GS_{sc} , as outlined below:

$$M(GS, GS_{sc}) = \begin{cases} 2.2, & \text{if } GS > GS_{sc} + 6 \\ 1.8, & \text{if } GS_{sc} + 3 < S \leq GS_{sc} + 6 \\ 1.4, & \text{if } GS_{sc} < S \leq GS_{sc} + 3 \\ 1.0, & \text{if } GS = GS_{sc} \\ \frac{1}{1.4}, & \text{if } GS_{sc} - 3 \leq GS < GS_{sc} \\ \frac{1}{1.8}, & \text{if } GS_{sc} - 6 \leq GS < GS_{sc} - 3 \\ \frac{1}{2.2}, & \text{if } GS < GS_{sc} - 6 \end{cases} \quad (1)$$

To clarify, the GS_{sc} value increments by one each second. At the start of a game, with an initial M value of 1.0, if a user's GS value exceeds 15 after the first 15 seconds, the M value is

adjusted based on both GS and GS_{sc} values. Conversely, if the GS value drops below 15, the M value decreases accordingly. However, if the GS value reaches precisely 15 after the 15 seconds, the M value remains at 1.0.

B. Experiment Design

A total of 14 participants, with an average age of 25.4 ± 2.35 years, comprising 10 males and 5 females, were enlisted from the university to participate in this study. Prior to the initiation of the experiments, researchers conducted a detailed briefing to inform participants with the study's objectives. After obtaining their agreement, participants were required to provide signed written consent. Subsequently, a trial session was conducted to familiarize participants with the Meta Quest Pro VR device's operation and virtual interactions within the game.

Participants were instructed to participate in the same game twice, once with the TC feature and once without (Non-TC or NTC). The order of these sessions was randomized, with half of the participants commencing with TC-based IVR exergaming followed by NTC-based IVR exergaming, and the other half initiating with NTC-based IVR exergaming. The total duration for NTC-based IVR exergaming was fixed at 300 seconds, while the duration for TC-based IVR exergaming varied based on participants' gaming performance. The study underwent a comprehensive review and obtained approval from the university's faculty research ethics committee, adhering to ethical review procedures outlined in the university's research conduct and research ethics guidelines [42].

Throughout the experiments, participants wore a Polar HR armband [43] to record HR data at a sampling rate of 1 Hz. Eyebrow movement (EM) and facial action unit data [44] were collected at a sampling rate of 10 Hz, starting 15 seconds after the game's commencement. Upon completing the experiments, each participant completed the IMI (see Section III-C).

C. IMI

The IMI serves as a multidimensional measurement tool designed to evaluate participants' subjective experiences concerning intrinsic motivation within the context of the experiments conducted in this study [16]. The standard IMI employed in this study comprises 22 scales relating to intrinsic motivation, which are further categorized into four sub-scales: enjoyment/interest, perceived choice, perceived competence, and pressure/tension. The enjoyment/interest sub-scale functions as a self-report measure of intrinsic motivation, while perceived choice and perceived competence are theorized to serve as positive predictors for both self-report and behavioral measures of intrinsic motivation. Conversely, pressure/tension is theorized to act as a negative predictor of intrinsic motivation.

D. Data Analysis

The emotional state of each participant was assessed using the Facial Action Coding System [44], which covers a comprehensive set of seven distinct emotions: "happy", "sad", "surprise", "fear", "anger", "disgust", and "contempt".

It is noteworthy that this study excluded the emotions of "surprise", "fear", "anger", "disgust", and "contempt" since the designed exergame was not specifically tailored to detect these emotions. A paired samples t -test was employed to evaluate the differences in emotional states, including heart rate (HR), eyebrow movement (EM), IMI, and gaming scores (GS), between conditions with TC and NTC. Additionally, the correlation between emotional state and HR, as well as between emotional state and EM, was computed and analyzed using Spearman's correlation test. Statistical Product and Service Solutions (SPSS) [45] was utilized as the statistical analysis tool.

IV. RESULTS AND DISCUSSION

The experimental findings revealed that the TC-based IVR exergaming was completed in a minimum duration of 143 seconds. Therefore, a sample size of 143 seconds was employed for the analysis of TC-related data. The mean emotional state μ was computed by considering the counts of detected emotions during gameplay, and the results are outlined in Table I.

The findings revealed that participants reported higher levels of happiness in the NTC condition ($\mu_{NTC}^{ES} = 4.71$, $STD_{NTC}^{ES} = 1.07$) compared to the TC condition ($\mu_{TC}^{ES} = 2.93$, $STD_{TC}^{ES} = 1.21$), with a $t = 3.617$ and $p = 0.003$. A greater quantity of sadness was observed in the NTC condition ($\mu_{NTC}^{ES} = 1.64$, $STD_{NTC}^{ES} = 1.22$) than in the TC condition ($\mu_{TC}^{ES} = 1.50$, $STD_{TC}^{ES} = 0.94$), with a $t = 0.322$ and $p = 0.752$. Meanwhile, neutral emotion was less prevalent in the NTC condition ($\mu_{NTC}^{ES} = 7.50$, $STD_{NTC}^{ES} = 1.74$) compared to the TC condition ($\mu_{TC}^{ES} = 9.64$, $STD_{TC}^{ES} = 1.45$), and this difference was statistically significant with a $t = -3.238$ and $p = 0.007$.

Overall, participants exhibited a greater range of happiness and sadness emotions in the NTC conditions, implying increased variability in emotional behavioral changes. Conversely, participants in the TC conditions predominantly displayed neutral emotions, indicating greater stability in emotional changes and suggesting a high adaptation of the TC feature in IVR exergaming.

However, upon thorough result analysis, no significant difference was observed in the changes of the EM data. Specifically, the EM values were lower in the NTC condition ($\mu_{NTC}^{EM} = 0.022$, $STD_{NTC}^{EM} = 0.038$) compared to the TC condition ($\mu_{TC}^{EM} = 0.025$, $STD_{TC}^{EM} = 0.035$), showing low significance for both TC and NTC conditions with a $t = -0.696$ and a $p = 0.487$. Conversely, a significant difference was found in the changes of HR, which exhibited lower values in the NTC condition ($\mu_{NTC}^{HR} = 107.73$, $STD_{NTC}^{HR} = 13.03$) than in the TC condition ($\mu_{TC}^{HR} = 110.06$, $STD_{TC}^{HR} = 12.58$), displaying significant differences for both TC and NTC conditions with a $t = -10.939$ and $p < 0.001$.

On the other hand, the mean GS per second in the NTC condition ($\mu_{NTC}^{GS} = 1.492$, $STD_{NTC}^{GS} = 0.197$) is notably lower than in the TC condition ($\mu_{TC}^{GS} = 1.520$, $STD_{TC}^{GS} = 0.232$), displaying significant differences for both TC and NTC conditions with a $t = -0.342$ and $p = 0.736$. In summary,

TABLE I: Statistical results of changes in emotional state (*ES*) under NTC and TC conditions

Emotion	μ_{NTC}	STD_{NTC}	μ_{TC}	STD_{TC}	t -value	p -value
Happy	4.71	1.07	2.93	1.21	3.617	0.003
Sad	1.64	1.22	1.50	0.94	0.322	0.752
Neutral	7.50	1.74	9.64	1.45	-3.238	0.007

TABLE II: Statistical results of changes in EM, HR and GS data under NTC and TC conditions

Data	μ_{NTC}	STD_{NTC}	μ_{TC}	STD_{TC}	t -value	p -value
EM	0.022	0.038	0.025	0.035	-0.696	0.487
HR	107.73	13.03	110.06	12.58	-10.939	< 0.001
GS	1.492	0.197	1.520	0.232	-0.342	0.736

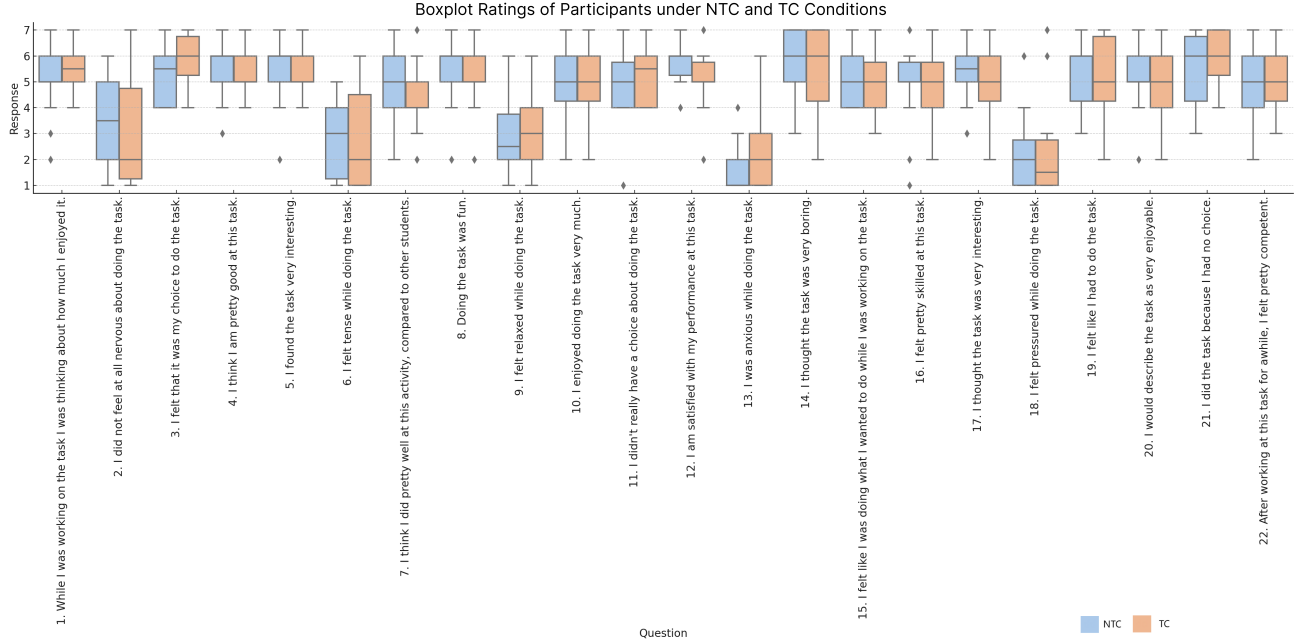


Fig. 3: Statistic results of IMI under NTC and TC conditions.

the results suggest that participants achieved better physical performance in the TC conditions, indicating the effectiveness of the individually tailored difficulty level. Table II provides a summary of the mean values of the changes in the EM, HR and GS data.

Meanwhile, a weak negative correlation between ES and EM data was identified for participants under the NTC condition, with a Spearman's correlation coefficient value of $c = -0.172$ and $p = 0.016$. Similarly, a weak negative correlation between ES and EM data was also observed for participants under the TC condition, with $c = -0.027$ and $p = 0.706$. Meanwhile, there was a negative correlation between ES and HR data for both TC ($c = -0.061$, $p = 0.396$) and NTC ($c = -0.074$, $p = 0.300$) conditions.

On the other hand, no significant difference is identified in the IMI for participant under the NTC and TC conditions ($t = 0.206$, $p = 0.837$), as shown in Fig. 3. Specifically, participants experienced game under NTC condition presented the mean rating of the enjoyment as $\mu_{NTC}^{EJ} = 5.34$ and $STD_{NTC}^{EJ} = 0.20$, perceived competence as $\mu_{NTC}^{PC} = 5.17$ and $STD_{NTC}^{PC} = 0.42$, perceived choice as $\mu_{NTC}^{PE} = 5.23$ and $STD_{NTC}^{PE} =$

0.26, and tension as $\mu_{NTC}^{TS} = 2.60$ and $STD_{NTC}^{TS} = 0.69$.

Meanwhile, participants who experienced the game with the TC feature reported mean ratings on the IMI in the aspect of enjoyment as $\mu_{TC}^{EJ} = 5.27$ with $STD_{TC}^{EJ} = 0.17$, perceived competence as $\mu_{TC}^{PC} = 4.94$ with $STD_{TC}^{PC} = 0.37$, perceived choice as $\mu_{TC}^{PE} = 5.44$ with $STD_{TC}^{PE} = 0.44$, and tension as $\mu_{TC}^{TS} = 2.64$ with $STD_{TC}^{TS} = 0.42$. Overall, the results revealed that mean ratings for enjoyment and perceived competence were higher under the NTC condition compared to the TC condition, while the mean ratings for perceived choice and tension were higher under the TC condition.

The study results revealed that adjusting the timer based on users' GS had a discernible impact on their emotional responses. The primary determinant of users' emotions in this context was the level of challenge presented by the exergame. In the NTC condition, where difficulty levels were not customized according to users' exergaming ability, participants exhibited a diverse range of emotional experiences. In contrast, TC-based gaming provided users with a more comfortable exergaming experience, associated with fewer extreme positive or negative emotions. The findings also corroborated the

principles of flow theory [46], indicating that optimizing the exergame challenge level within a comfort zone could sustain user engagement without causing frustration or boredom.

Throughout the study, it was observed that as the timer speed increased with improved GS, participants tended to achieve even higher GS. Consequently, during TC adaptation intervals, participants felt compelled to enhance their movement speed, resulting in elevated HRs due to increased focus. This phenomenon aligns with the finding of [47], which elucidated that heightened HR correlates with increased oxygenated blood circulation during periods of intensified physical activity. Conversely, participants with more experience in IVR gaming achieved high GS in less time, but their overall experience remained consistent when exposed to TC-based exergaming. In contrast, less experienced IVR exergaming participants performed better with a slower timer change, exhibiting more comfortable gaming experiences.

Moreover, during the NTC-based exergaming experiments, it was noted that participants who perceived their performances as below expectations tended to experience a sense of diminished control over the game progress. This resulted in increased emotional fluctuations and a decline in gaming performance. In conclusion, this study proposed that neutral emotion serves as a promising satisfactory indicator for the design of the DDA mechanism in IVR gaming.

V. CONCLUSION

This study examined changes in participants' emotional behavior during IVR exergaming under two conditions: games with the TC feature and without the TC feature (NTC). The results revealed that participants in the exergame with the TC feature, a popular DDA mechanism, exhibited a higher frequency of neutral emotions and improved in-game performance. Conversely, participants in the exergame without the TC feature displayed more variability in emotions, suggesting that the difficulty level provided diverse experiences, lacking personalized adjustments for individual needs. Moreover, the DDA mechanism with the TC feature in IVR exergaming, providing personalized challenge adjustments, could lead to more neutral emotions. These preliminary findings lay the groundwork for future research into the development of autonomous DDA mechanism-enabled IVR exergaming, which takes into account changes in users' physical and mental states.

The study has certain limitations, such as a relatively small sample size, potentially limiting the generalizability of the results. Additionally, exploring different exergames with participants from diverse demographics is necessary for a comprehensive understanding of emotional behavior responses to DDA. Future research should consider investigating various DDA mechanisms beyond TC and aim to design an optimized autonomous DDA mechanism based on both the physical and mental conditions of IVR exergaming users.

REFERENCES

- [1] V. Benzing and M. Schmidt, "Exergaming for children and adolescents: strengths, weaknesses, opportunities and threats," *Journal of clinical medicine*, vol. 7, no. 11, p. 422, 2018.
- [2] R. Dörner, S. Göbel, W. Effelsberg, and J. Wiemeyer, *Serious games*. Springer, 2016.
- [3] J. Bolton, M. Lambert, D. Lirette, and B. Unsworth, "Paperdude: a virtual reality cycling exergame," in *CHI'14 Extended Abstracts on Human Factors in Computing Systems*, 2014, pp. 475–478.
- [4] A. Szpak, S. C. Michalski, and T. Loetscher, "Exergaming with beat saber: an investigation of virtual reality aftereffects," *Journal of Medical Internet Research*, vol. 22, no. 10, p. e19840, 2020.
- [5] J. F. V. Castillo, M. F. M. Vega, J. E. M. Cardona, D. Lopez, L. Quiñones, O. A. H. Gallo, and J. F. Lopez, "Design of virtual reality exergames for upper limb stroke rehabilitation following iterative design methods: Usability study," *JMIR Serious Games*, vol. 12, no. 1, p. e48900, 2024.
- [6] C. V. Sousa, J. Hwang, R. Cabrera-Perez, A. Fernandez, A. Misawa, K. Newhook, and A. S. Lu, "Active video games in fully immersive virtual reality elicit moderate-to-vigorous physical activity and improve cognitive performance in sedentary college students," *Journal of Sport and Health Science*, vol. 11, no. 2, pp. 164–171, 2022.
- [7] S. Finkelstein, A. Nickel, Z. Lipps, T. Barnes, Z. Wartell, and E. A. Suma, "Astrojumper: Motivating exercise with an immersive virtual reality exergame," *Presence: Teleoperators and Virtual Environments*, vol. 20, no. 1, pp. 78–92, 2011.
- [8] E. G. Murray, D. L. Neumann, R. L. Moffitt, and P. R. Thomas, "The effects of the presence of others during a rowing exercise in a virtual reality environment," *Psychology of Sport and Exercise*, vol. 22, pp. 328–336, 2016.
- [9] R. Hunicke, "The case for dynamic difficulty adjustment in games," in *Proceedings of the 2005 ACM SIGCHI International Conference on Advances in computer entertainment technology*, 2005, pp. 429–433.
- [10] S. Xue, M. Wu, J. Kolen, N. Aghdaie, and K. A. Zaman, "Dynamic difficulty adjustment for maximized engagement in digital games," in *Proceedings of the 26th International Conference on World Wide Web Companion*, 2017, pp. 465–471.
- [11] O. S. Ajani and R. Mallipeddi, "Pareto-based dynamic difficulty adjustment of a competitive exergame for arm rehabilitation," *International Journal of Human-Computer Studies*, p. 103100, 2023.
- [12] M. Pezzera and N. A. Borghese, "Dynamic difficulty adjustment in exergames for rehabilitation: a mixed approach," in *2020 IEEE 8th International Conference on Serious Games and Applications for Health (SeGAH)*. IEEE, 2020, pp. 1–7.
- [13] L. Kruse, S. Karaosmanoglu, S. Rings, and F. Steinicke, "Evaluating difficulty adjustments in a vr exergame for younger and older adults: Transferabilities and differences," in *Proceedings of the 2022 ACM Symposium on Spatial User Interaction*, 2022, pp. 1–11.
- [14] P. Ramasamy, G. Renganathan, and Y. Kurita, "Force feedback-based gamification: Performance validation of squat exergame using pneumatic gel muscles and dynamic difficulty adjustment," *IEEE Robotics and Automation Letters*, 2023.
- [15] A. Bouatrous, A. Meziane, N. Zenati, and C. Hamitouche, "A new adaptive vr-based exergame for hand rehabilitation after stroke," *Multimedia Systems*, vol. 29, no. 6, pp. 3385–3402, 2023.
- [16] R. Ryan and E. Deci, "Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being," *The American psychologist*, vol. 55, pp. 68–78, 01 2000.
- [17] W. Peng, J.-H. Lin, and J. Crouse, "Is playing exergames really exercising? a meta-analysis of energy expenditure in active video games," *Cyberpsychology, Behavior, and Social Networking*, vol. 14, no. 11, pp. 681–688, 2011.
- [18] J. Sween, S. F. Wallington, V. Sheppard, T. Taylor, A. A. Llanos, and L. L. Adams-Campbell, "The role of exergaming in improving physical activity: a review," *Journal of Physical Activity and Health*, vol. 11, no. 4, pp. 864–870, 2014.
- [19] M. Farrow, C. Lutteroth, P. C. Rouse, and J. L. Bilzon, "Virtual-reality exergaming improves performance during high-intensity interval training," *European journal of sport science*, vol. 19, no. 6, pp. 719–727, 2019.
- [20] K.-T. Huang, "Exergaming executive functions: An immersive virtual reality-based cognitive training for adults aged 50 and older," *Cyberpsychology, Behavior, and Social Networking*, vol. 23, no. 3, pp. 143–149, 2020.
- [21] A. J. Seyderhelm and K. Blackmore, "Systematic review of dynamic difficulty adaption for serious games: the importance of diverse approaches," *Available at SSRN 3982971*, 2021.
- [22] A. Denisova and P. Cairns, "First person vs. third person perspective in digital games: do player preferences affect immersion?" in *Proceedings*

of the 33rd annual ACM conference on human factors in computing systems, 2015, pp. 145–148.

- [23] J. T. Alexander, J. Sear, and A. Oikonomou, “An investigation of the effects of game difficulty on player enjoyment,” *Entertainment computing*, vol. 4, no. 1, pp. 53–62, 2013.
- [24] J. Suaza, E. Gamboa, and M. Trujillo, “A health point-based dynamic difficulty adjustment strategy for video games,” in *Joint International Conference on Entertainment Computing and Serious Games*. Springer, 2019, pp. 436–440.
- [25] A. Rietveld, S. Bakkes, and D. Roijers, “Circuit-adaptive challenge balancing in racing games,” in *2014 IEEE Games Media Entertainment*. IEEE, 2014, pp. 1–8.
- [26] S. M. De la Cruz, S. S. Portocarrero, and P. Shiguihara-Juárez, “Adaptive serious game as a learning approach for microbiology,” in *2018 Congreso Internacional de Innovación y Tendencias en Ingeniería (CONIITI)*. IEEE, 2018, pp. 1–6.
- [27] L. Tahai, J. R. Wallace, C. Eckhardt, and K. Pietroszek, “Scalebridge: Design and evaluation of adaptive difficulty proportional reasoning game for children,” in *2019 11th International Conference on Virtual Worlds and Games for Serious Applications (VS-Games)*. IEEE, 2019, pp. 1–4.
- [28] K. Spiel, S. Bertel, and F. Kayali, “Adapting gameplay to eye movements—an exploration with tetris,” in *Extended Abstracts of the Annual Symposium on Computer-Human Interaction in Play Companion Extended Abstracts*, 2019, pp. 687–695.
- [29] M. Ninaus, K. Tsarava, and K. Moeller, “A pilot study on the feasibility of dynamic difficulty adjustment in game-based learning using heart-rate,” in *Games and Learning Alliance: 8th International Conference, GALA 2019, Athens, Greece, November 27–29, 2019, Proceedings 8*. Springer, 2019, pp. 117–128.
- [30] L. Reidy, D. Chan, C. Nduka, and H. Gunes, “Facial electromyography-based adaptive virtual reality gaming for cognitive training,” in *Proceedings of the 2020 International Conference on Multimodal Interaction*, 2020, pp. 174–183.
- [31] A. Stein, Y. Yotam, R. Puzis, G. Shani, and M. Taieb-Maimon, “Eeg-triggered dynamic difficulty adjustment for multiplayer games,” *Entertainment computing*, vol. 25, pp. 14–25, 2018.
- [32] K. Mikami, K. Kondo *et al.*, “Adaptable game experience based on player’s performance and eeg,” in *2017 Nicograph International (NicoInt)*. IEEE, 2017, pp. 1–8.
- [33] G. Chanel and P. Lopes, “User evaluation of affective dynamic difficulty adjustment based on physiological deep learning,” in *International Conference on Human-Computer Interaction*. Springer, 2020, pp. 3–23.
- [34] W. S. Ravyse, A. Seugnet Blignaut, V. Leendertz, and A. Woolner, “Success factors for serious games to enhance learning: a systematic review,” *Virtual Reality*, vol. 21, pp. 31–58, 2017.
- [35] E. Galy, M. Cariou, and C. Mélan, “What is the relationship between mental workload factors and cognitive load types?” *International journal of psychophysiology*, vol. 83, no. 3, pp. 269–275, 2012.
- [36] T. Huber, S. Mertes, S. Rangelova, S. Flutura, and E. André, “Dynamic difficulty adjustment in virtual reality exergames through experience-driven procedural content generation,” in *2021 IEEE Symposium Series on Computational Intelligence (SSCI)*. IEEE, 2021, pp. 1–8.
- [37] D. S. Lora Ariza, A. A. Sánchez-Ruiz, and P. A. González-Calero, “Towards finding flow in tetris,” in *Case-Based Reasoning Research and Development: 27th International Conference, ICCBR 2019, Otzenhausen, Germany, September 8–12, 2019, Proceedings 27*. Springer, 2019, pp. 266–280.
- [38] D. Hooshyar, M. Pedaste, Y. Yang, L. Malva, G.-J. Hwang, M. Wang, H. Lim, and D. Delev, “From gaming to computational thinking: An adaptive educational computer game-based learning approach,” *Journal of Educational Computing Research*, vol. 59, no. 3, pp. 383–409, 2021.
- [39] Unity real-time development platform. [Online]. Available: <https://www.unity.com>
- [40] Face tracking for movement sdk for unity. [Online]. Available: <https://developer.oculus.com/documentation/unity/move-face-tracking/face-blendshapes>
- [41] A. Denisova and P. Cairns, “Adaptation in digital games: the effect of challenge adjustment on player performance and experience,” in *Proceedings of the 2015 Annual Symposium on Computer-Human Interaction in Play*, 2015, pp. 97–101.
- [42] University of nottingham ningbo china research ethics. [Online]. Available: <https://www.nottingham.edu.cn/en/research-and-business/ethics.aspx>
- [43] Heart rate sensors. [Online]. Available: <https://www.polar.com/us-en/products/heart-rate-sensors/>
- [44] P. Ekman and W. V. Friesen, “Facial action coding system,” *Environmental Psychology & Nonverbal Behavior*, 1978.
- [45] Ibm spss statistics. [Online]. Available: <https://www.ibm.com/cnzh/products/spss-statistics>
- [46] M. Csikszentmihalyi, S. Abuhamdeh, and J. Nakamura, “Flow,” *Handbook of competence and motivation*, pp. 598–608, 2005.
- [47] D. U. Silverthorn, W. C. Ober, C. W. Garrison, A. C. Silverthorn, and B. R. Johnson, *Human physiology: an integrated approach*. Pearson Education Indianapolis, IN, 2013, vol. 3.

Enhancing User Motivation in Extended Reality-Assisted Rehabilitation: A Comprehensive Systematic Review

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Abstract—This review paper thoroughly investigates techniques aimed at improving users’ motivation in the context of Extended Reality (XR)-assisted rehabilitation. The potential of XR to revolutionize rehabilitation practices is explored, with applications spanning musculoskeletal rehabilitation, neurological rehabilitation, cognitive rehabilitation, and psychological therapy. These applications are tailored to address a range of medical conditions, including stroke, amputations, brain injuries, and mental health disorders. The integration of XR in rehabilitation brings significant benefits such as immersive environments, real-time feedback, and personalized interventions. However, the challenge of maintaining patient motivation during XR-assisted rehabilitation adversely impacts treatment adherence and overall effectiveness. This paper presents a comprehensive systematic review of 28 carefully selected studies that investigate techniques for enhancing users’ motivation in the context of XR-assisted rehabilitation. These studies cover diverse populations, ranging from stroke survivors and individuals with multiple sclerosis to those with limb injuries, transfemoral amputation, and the elderly. The motivation-enhancing strategies identified in these studies include gamification, social interaction, and feedback mechanisms. While the results of these studies are promising, the review underscores the need for larger-scale research efforts to strengthen the level of evidence. In summary, this review paper offers valuable insights for researchers, healthcare professionals, and developers involved in the design of effective and engaging XR interventions. The objective is to improve rehabilitation outcomes and enhance the overall rehabilitation experience for individuals dealing with physical or cognitive impairments.

Index Terms—Augmented and virtual reality, extended reality, games, human-computer interface, rehabilitation.

I. INTRODUCTION

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EXTENDED reality (XR), an inclusive concept comprising virtual reality (VR), augmented reality (AR), and mixed reality (MR) [1], has garnered substantial attention across diverse domains, notably in healthcare [2]. XR technologies have demonstrated remarkable potential in revolutionizing rehabilitation practices by offering immersive and interactive experiences tailored to individuals with physical or cognitive impairments [3]–[5]. A plethora of studies has underscored the advantages of XR over conventional rehabilitation approaches, emphasizing its capacity to enhance patient engagement [6]–[8].

In the realm of rehabilitation, XR has found extensive utility across diverse domains to address a spectrum of therapeutic needs. XR interventions have been applied in areas such as musculoskeletal rehabilitation [9], neurological rehabilitation [3], cognitive rehabilitation [10], and psychological therapy [11]. These applications have targeted a range of conditions, including stroke [12], spinal cord injury [13], amputations [14], brain injury [15], and mental health disorders [16]. The versatility of XR technology permits personalized interventions that address specific impairments and challenges faced by individuals undergoing rehabilitation [17].

The integration of XR into rehabilitation offers several advantages. XR provides a highly immersive environment that enables users to engage in realistic simulations and activities [18]. This heightened realism can enhance the sense of presence and embodiment, enabling individuals to practice functional tasks and movements within a secure and controlled setting [19], [20]. In addition, XR can deliver real-time and customized feedback, enabling users to monitor their progress and adjust their performance accordingly. This feedback mechanism promotes self-awareness and motivates individuals to actively participate in their rehabilitation process [21]–[23]. Lastly, XR interventions can be personalized and adapted to align with individual needs, preferences, and levels of ability. The flexibility inherent to XR technology allows for tailored interventions that optimize outcomes and enhance treatment effectiveness [17], [24].

Nevertheless, despite the potential advantages of XR-assisted rehabilitation, patients frequently grapple with motivational challenges during their treatment [25]. Fostering and sustaining motivation for consistent engagement in rehabilitation exercises can pose a considerable obstacle, impacting treatment adherence and, ultimately, the intervention’s efficacy [26]. Gaining a comprehensive understanding of the factors

contributing to this motivational deficit is imperative for crafting strategies to address and surmount it.

This review paper endeavors to investigate and expound upon methods for augmenting user motivation in the context of XR-assisted rehabilitation. Through the systematic aggregation and analysis of pertinent evidence, this study aspires to pinpoint pivotal factors that amplify users' motivation and participation. By tackling this critical challenge, this study aspires to make a meaningful contribution to the advancement of more efficacious and patient-centered XR-assisted rehabilitation programs. In particular, the contributions of this review study encompass:

- A comprehensive synthesis and analysis of the extant literature in the realm of user motivation within XR-assisted rehabilitation.
- An exhaustive examination of strategies to enhance user motivation in XR-assisted rehabilitation, encompassing diverse approaches such as gamification, social interaction, feedback mechanisms, and other modalities.
- An exploration of prospective avenues for future research, including harnessing the potential of artificial intelligence (AI) for dynamic adjustments in game difficulty and enriched narratives, delving into the realm of human-AI collaboration, and contemplating the application of neurofeedback via brain-computer interfaces (BCIs).

II. METHODS

In accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [26], this systematic review aimed to investigate techniques for enhancing user motivation in XR-assisted rehabilitation. The review's objective was to compile and evaluate existing evidence within this research domain.

A. Eligibility Criteria

The identification of suitable articles followed the PICO framework [27]. The population of interest (P) included individuals of all ages and genders undergoing XR-assisted rehabilitation for physical or cognitive impairments. The intervention (I) was XR, while comparison groups (C) encompassed those receiving standard care or non-XR rehabilitation interventions. Primary outcomes of interest (O) centered on users' motivation improvement during XR-assisted rehabilitation, with secondary outcomes covering treatment adherence, engagement, and user satisfaction. To maintain a focused search, review papers, case reports, study protocols, conference papers, and non-English language papers were excluded from consideration. However, there were no restrictions on publication years, as XR is a relatively recent addition to healthcare research.

B. Search Strategy

For this review study, information sources included the Web of Science and PubMed databases. Web of Science offers a wide range of high-quality academic literature across various fields, while PubMed provides information relevant to

rehabilitation, crucial for this review. The research question guiding the systematic review was: "How can users' motivations be improved in XR-assisted rehabilitation?" The search strategy underwent several refinements, including the removal of redundant keywords to ensure the most relevant results. The applied search approach was as follows: ("virtual reality" OR "augmented reality" OR "extended reality") AND ("rehabilitation" OR "therapy") AND ("motivation" OR "inspiration").

C. Study Selection

All search results from Web of Science and PubMed were managed using the Zotero reference management tool. Duplicate records were initially removed before the screening process. Subsequently, irrelevant papers were excluded based on an initial assessment of their titles and abstracts. After full-text screening, studies that met the predetermined eligibility criteria were included in the review. A comprehensive search of PubMed and Web of Science databases yielded a total of 1522 published articles. After removing duplicate records, 1155 articles remained, considered potentially relevant based on an initial review of titles and abstracts. Subsequently, 103 full-text articles were assessed for eligibility criteria. Ultimately, 28 publications were selected and included in the systematic review. The study selection process is illustrated in the flowchart presented in Fig. 1.

III. RESULTS

A. Study Characteristics

The 28 studies included in this review were published between 2019 and 2023. These studies featured diverse sample sizes, ranging from five to 113 participants. The study populations were composed of healthy adults [12], [17], [28]–[37] and individuals with various medical conditions, including spatial neglect [38], physical disabilities [39], stroke [12], [24], [40]–[44], upper limb injuries [45], [46], multiple sclerosis [24], [47], essential hypertension [48], neuromuscular disorders [46], transfemoral amputation [14], and attention disorders [49]. Twelve of these studies employed immersive-based VR (IVR) [10], [12], [17], [28]–[33], [35], [42], [45], another twelve used non-immersive-based VR (NIVR) [24], [36], [37], [39]–[41], [43], [44], [46], [47], [49], [50], two incorporated augmented reality (AR) [34], [38], and two employed mixed reality (MR) techniques [14], [48]. These studies covered a wide range of interventions aimed at enhancing users' motivation in XR-assisted rehabilitation.

B. Results of Individual Studies

The comprehensive summary outlining the rehabilitation modalities, participants, interventions, and core techniques to improve user motivation in each study included in this review can be found in Table I.

1) *Immersive-Based VR (IVR)*: Twelve studies were carried out in immersive VR scenarios. Ventura *et al.* [30] introduced an innovative method for remote identification and classification of human movements, particularly in the context of bimanual training in VR for motor performance assessment

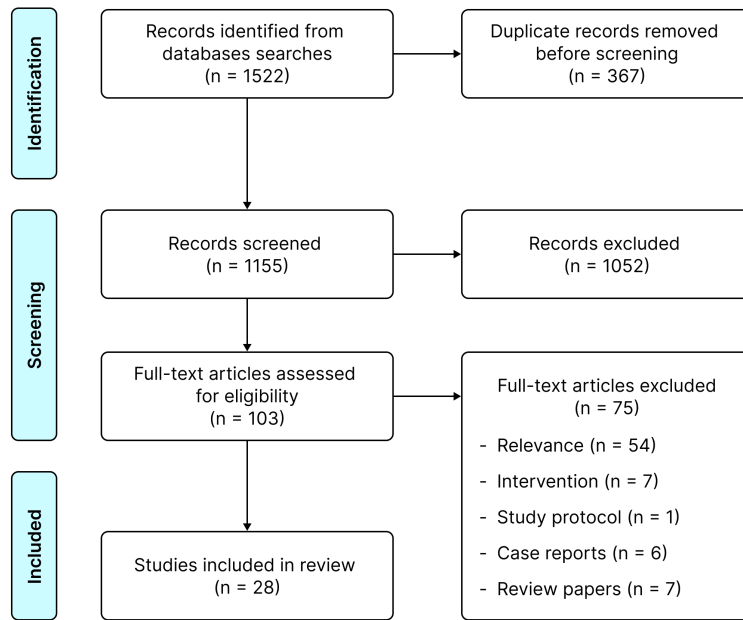


Fig. 1. PRISMA-based study selection flowchart.

TABLE I
OVERVIEW OF REHABILITATION MODALITIES, PARTICIPANTS, INTERVENTIONS, AND FUNDAMENTAL TECHNIQUES FOR ENHANCING USER MOTIVATION

Related Studies	Rehabilitation Modalities	Participants	Interventions	Techniques for Enhancing Users Motivation
[38]	Spatial Neglect	Patients	AR	Games
[39]	Physical Disabilities	Patients	NIVR	Games
[24]	Stroke and Multiple Sclerosis	Patients	NIVR	Games
[30]	Bimanual Movements	Normal Users	IVR	Games
[48]	Hypertension	Patients	MR	Games
[28]	Amblyopia	Normal Users	IVR	Games
[17]	Upper Limb Problems	Normal Users	IVR	Games
[40]	Stroke	Patients	NIVR	Games
[45]	Upper Limb Injuries	Patients	IVR	Games
[46]	Neuromuscular Disorders	Patients	NIVR	Games
[31]	Upper Limb Exercises	Normal Users	IVR	Games
[29]	Hand Motor Problems	Patients	IVR	Games
[10]	Stroke	Patients	IVR	Games
[41]	Stroke	Patients	NIVR	Games
[32]	Stroke	Normal Users	IVR	Games
[42]	Stroke	Patients	IVR	Games
[33]	Neurorehabilitation	Normal Users	IVR	Games
[43]	Stroke	Patients	NIVR	Games
[44]	Stroke	Patients	NIVR	Games
[35]	Full Body Exercises	Normal Users	IVR	Games & Social Interaction
[50]	Stroke	Patients	NIVR	Games & Social Interaction
[36]	Frailty Improvement	Normal Users	NIVR	Games & Social Interaction
[49]	Attention Deficit and Hyperactivity Disorder	Patients	NIVR	Games, Feedback Mechanism
[14]	Transfemoral Amputation	Patients	MR	Feedback Mechanism
[12]	Stroke	Patients	IVR	Feedback Mechanism
[34]	Upper Limb Injuries	Normal Users	AR	Games, Emotion Monitoring
[37]	Developmental Dysplasia of the Hip (DDH)	Normal Users	NIVR	Games, Persuasive System
[47]	Multiple Sclerosis	Patients	NIVR	Games, Virtual Objects Interaction

during telerehabilitation. Khaleghi *et al.* [28] explored the application of VR-assisted gamified binocular exercises to treat amblyopia in adults, focusing on enhancing visual acuity, motivation, and engagement. Izountar *et al.* [17] introduced the VR-PEER adaptive exergame system, which takes into account patient emotions during rehabilitation exercises, potentially improving rehabilitation outcomes. Elor *et al.* [45] investigated

the effects of a two-month VR physical therapy program for upper-extremity motions, emphasizing user engagement and progress evaluation. Chen *et al.* [31] studied the acceptance of upper limb exercises through IVR games, with a focus on the roles of age, self-efficacy, and technology acceptance model (TAM) constructs.

Chatterjee *et al.* [10] conducted a double-blind randomized

controlled trial (RCT) using a custom VR environment known as VIRTUE for cognitive rehabilitation in stroke patients, incorporating gamification techniques to enhance motivation. Ziadeh *et al.* [32] explored the influence of embodiment in VR on the sense of agency during a binary motor imagery task with a BCI, discussing its impact on agency, ownership, and frustration in MI-BCI tasks. Moan *et al.* [42] conducted a qualitative pilot study to delve into the experiences of stroke survivors and clinicians with a fully immersive VR treadmill exergame for stroke rehabilitation, revealing that participants found motivation in the game through safety, engagement, and achievement.

Knobel *et al.* [33] developed a visual search task within an immersive VR setting called the “bird search task”, with the game’s difficulty automatically adapting based on each participant’s performance. Shah *et al.* [35] introduced a collaborative exergame in social VR for the elderly’s rehabilitation, underscoring the role of collaboration in motivating exercise and promoting social interaction. Dong *et al.* [12] developed a haptic-feedback VR system to enhance the assessment of upper extremity motor function in both healthy individuals and post-stroke patients, finding it to be reliable, motivational, and enjoyable. Juan *et al.* [29] presented a VR application for motor rehabilitation that utilizes hand gestures as the means of interaction, with most participants preferring this natural interaction method.

2) *Non-Immersive-Based VR (NIVR)*: Twelve studies were conducted in non-immersive VR scenarios. Eckert *et al.* [39] performed an observational case study involving six children with diverse physical disabilities who received physical therapy at school along with the use of a fully configurable exergame integrated into the Blexer environment. This study underscored the importance of correctly configuring the game’s difficulty parameters. Wiskerke *et al.* [24] utilized the Rasch model to create a hierarchy of difficulty levels for VR balance exercises, a valuable resource for individuals with stroke and multiple sclerosis with varying balance abilities. Herne *et al.* [40] conducted a multiple case study to examine the engagement of stroke survivors with a desktop VR-based serious game for upper limb rehabilitation, focusing on the role of feedback as a crucial design factor.

Eckert *et al.* [46] presented a modular approach to generic exergame design, combining custom physical exercises within a meaningful and motivating narrative, albeit suggesting the need for further fine-tuning of the game’s 3D exercises to enhance fun and motivation. Allegue *et al.* [41] conducted a multiple case study involving three stroke survivors and their clinicians to investigate the VirTele intervention, a remote rehabilitation program combining VR exergames and telerehabilitation to improve upper extremity deficits in stroke survivors. Lehmann *et al.* [43] employed the YouGrabber (YG) VR program for upper limb practice in chronic stroke patients, exploring their experiences and improvements in motor function. Fluet *et al.* [44] studied the ability of stroke survivors to use a home virtual rehabilitation system (HoVRS) and the impact of motivational enhancement techniques on motivation, adherence, and motor function improvement.

Navarro *et al.* [50] conducted a randomized controlled trial

to investigate the effectiveness and motivating abilities of an intervention designed to address attention deficits in individuals post-stroke, utilizing paper-and-pencil tasks and interactive, computerized, multi-touch exercises, either non-competitively or competitively, for individuals with chronic stroke. Arlati *et al.* [36] assessed the feasibility and acceptability of the Social-Bike application for physical and cognitive training in elderly individuals, with promising outcomes.

Skalski *et al.* [49] carried out a study to evaluate the effects of hemoencephalographic (HEG) biofeedback (BFB) with VR on attention deficits in children with ADHD, concluding that HEG BFB with VR has a beneficial effect in treating attention. Sanpablo *et al.* [37] applied the Persuasive System Design (PSD) model to analyze and develop exergame systems for gait rehabilitation in pediatric patients, integrating persuasive elements like adjusting treadmill speed, auditory feedback, and game challenges into a video gaming and VR biofeedback system for treadmill gait rehabilitation. Cikajlo *et al.* [47] examined the impact of virtual object size and bounce on motor performance and motivation in individuals with multiple sclerosis, offering insights relevant to occupational therapy.

3) *AR*: Two studies were conducted in AR scenarios. Stammler *et al.* [38] developed an AR-based app, Negami, designed for spatial neglect treatment. This app combines visual exploration training with active, contralesionally oriented head, eye, and trunk rotations. The natural interaction with the physical environment during playful tasks minimized side effects associated with cybersickness and notably increased patients’ motivation. Rodrigues *et al.* [34] created a pre-prototype of a rehabilitation system that integrates a collaborative robot with AR equipment to provide a customized and gamified rehabilitation experience for patients.

4) *MR*: Two studies took place in MR scenarios. Stamm *et al.* [48] examined the usability and acceptance of exergames in older hypertensive patients, employing various training types within a simulated MR setting. The results indicated that usability and acceptance were not linked to the type of training when using MR exergames. Lancere *et al.* [14] utilized MR technologies and wireless sensors to enhance muscle engagement in lower extremity amputee rehabilitation, focusing on the effectiveness of verbal and visual MR feedback in improving muscle activation and motivation.

IV. DISCUSSION

The incorporated studies have delved into a wide array of applications for XR, spanning from the enhancement of muscle engagement in lower extremity amputee rehabilitation to the augmentation of upper extremity motor function assessments. The consistent findings across these studies have underscored the presence of positive outcomes. These encompass amplified motivation, heightened self-awareness, increased engagement, and the advancement of both physical and cognitive functions. Furthermore, the research highlights the pivotal role of several techniques in augmenting user experience and motivation within these XR-based applications. These techniques encompass gamification, social interaction, sensory feedback, facial emotion recognition, and the integration of persuasive systems.

TABLE II
SUMMARY OF TECHNIQUES TO ENHANCE USER MOTIVATION IN XR-ASSISTED REHABILITATION

Techniques	Mechanisms	Descriptions
Gamification	Game Difficulty Level	• Alignment of different game levels with therapy [28]. • Progressive increase of the difficulty level [10]. • Dynamic adjustment of difficulty levels based on users' performance [33]. • Fine-tuning of the scoring mechanisms and integration of visually appealing graphics that align with the changing difficulty [44]. • Application of the Rasch model to identify the difficulty levels [24]. • Dynamic adjustment of difficulty level based on emotion recognition [17].
	Narrative	• Progression of the engaging storyline [46]. • Motivating through relatedness, competence, and autonomy based on self-determination theory [45]. • Incorporating motivational interviewing techniques consistent with the principles of self-determination theory [41].
	Embodiment of Avatars	• Most users prefer interacting with their hands [29]. • Avatar hands provide a more anthropomorphic and relatable representation, significantly increasing users' sense of ownership compared to abstract blocks [32].
	Virtual Objects Interaction	• Manipulating the size and behavior of virtual objects as a mechanism for proximal and fine motor performance evaluation [47].
	Persuasive System	• Integration of persuasive elements into exergames [37].
Social Interaction	Citizen Science	• Combination of citizen science and gamification [30].
	Collaboration	• Implementing social collaboration in XR-assisted rehabilitation that encourages users to perform rehabilitation with others instead of individually [35], [36].
	Competition	• Enhancing the effectiveness and enjoyment of rehabilitation but potentially introduce negative effects if exercises are overdone [50].
Feedback	Visual-Audio	• Provision of real-time verbal and visual feedback through MR [14].
	Haptic	• Usage of haptic-feedback in VBBT system [12].
	Biofeedback	• Monitoring cerebral blood oxygenation [49].

Nevertheless, despite the promise and favorable results, it remains imperative to conduct further research. This additional investigation aims to validate and refine these technologies, surmount challenges such as stress levels and limitations in sample size, and ultimately cultivate more effective and captivating XR-based interventions across diverse populations. For a more detailed overview of these techniques, please refer to Table II.

A. Gamification

The strategic incorporation of gamification has emerged as a promising approach to augment user motivation, as evidenced by various studies [34], [40], [42], [48]. This technique revolves around several key elements, which encompass the difficulty level, narrative, embodiment of avatars, interaction with virtual objects, persuasive systems, and the adoption of citizen science principles.

1) Game Difficulty Level: The variable levels of game difficulty within XR-assisted rehabilitation applications exert a profound influence on individual motivation. In this context, difficulty extends beyond the physical demands of exercises to encompass the complexity of virtual environments, the intricacy of presented tasks, and the required cognitive engagement. Lower difficulty levels feature simpler and repetitive exercises, promoting basic muscle movement or cognitive function. Conversely, higher difficulty levels involve more challenging and interactive scenarios demanding advanced motor skills and intense cognitive processing.

Khaleghi *et al.* [28] implemented a simple game design based on Jafari *et al.*'s study [51], which proposed that less challenging games are more effective as therapeutic tools, particularly for adult users. Nonetheless, evaluations indicated areas for improvement, such as incorporating a storyline, introducing diverse mini-games to accommodate users' varying eye conditions, and implementing different levels aligned with the

therapeutic components of the study. This approach can prevent monotony and foster increased patient interaction, thereby enhancing effectiveness. Chatterjee *et al.* [10] discovered that gamification techniques can heighten patients' motivation by progressively increasing the difficulty level of tasks over time. The difficulty in their study was modified based on the number of ingredients in a recipe. Patients first tackled simpler tasks before progressing to more challenging ones. Stammeler *et al.* [38] found that training at the highest defined difficulty level was perceived as challenging but not frustrating. Knobel *et al.* [33] recommended automatic adjustments of difficulty levels based on users' performance in the previous round. When altering the difficulty level of games, it is crucial to consider its overall impact on the gaming experience. Fine-tuning scoring mechanisms and integrating visually appealing graphics that align with changing difficulty levels are instrumental in maintaining user engagement and motivation [44].

The utilization of the Rasch model has proven effective in assessing the balance scale items, ordering them from easily executable by all participants to difficult items beyond the capability of all participants [52]. Wiskerke *et al.* [24] successfully employed this model to establish a hierarchy of VR balance exergames for individuals with neurological disorders, such as stroke and multiple sclerosis, who possess low to moderate balance abilities. This approach aids in aligning VR exercises with patients' abilities, addressing the common challenge of underchallenging experiences in VR exergames. By integrating this hierarchy with the Berg Balance Scale, a widely used clinical assessment tool, clinicians can make more informed decisions when designing individualized rehabilitation programs for patients with neurological disorders.

While personalized games have adapted to individual characteristics and needs, they have primarily focused on the progression of game difficulty rather than considering patients' emotional states. This limitation is significant because it's challenging to discern a patient's emotional state during gameplay. Patients may experience stress, fatigue, or anxiety without the game's awareness, inadvertently leading to negative performance outcomes. To mitigate this, Izountar *et al.* [17] proposed a dynamic exergame that considers patients' facial emotions. By monitoring their facial expressions, the game dynamically adjusts its difficulty level, tailoring the gaming experience to individual patients. This personalized approach not only boosts motivation but also ensures the game's difficulty aligns with the patient's emotional state, potentially improving outcomes and the overall patient experience.

2) *Narrative*: Incorporating engaging game narrative into XR-assisted rehabilitation has emerged as a potent tool for enhancing user engagement and intrinsic motivation. Players find motivation in the story and the pursuit of high scores [46]. Previous research has shown that participating in acts of helping behavior can yield diverse rewards, encompassing material, social, and self-related benefits [53]. Caring and helping relationships are strong factors in promoting long-term engagement [54]. Elor *et al.* [45] extended these ideas by incorporating them into an Immersive Virtual Reality (IVR) game called "Project Butterfly." The game encouraged players

to help and protect a virtual butterfly, fostering a sense of attachment and belonging as its guardian. Drawing on self-determination theory [55], the game aimed to intrinsically motivate users through relatedness, competence, and autonomy. This approach fosters an emotional connection and motivates users to engage in physical activity.

3) *Embodiment of Avatars*: Hand interaction has gained popularity due to its intuitive and natural user experience. Juan *et al.* compared two types of interactions: using hands versus controllers. Both groups successfully completed exercises, with no significant differences in the number of attempts between the two interaction types. While the controller group completed exercises more quickly, a majority of participants (78.5%) preferred interacting with their hands. This finding suggests that combining user-hand interaction with standalone headsets can enhance motivation, gain acceptance among motor rehabilitation patients, and facilitate exercise rehabilitation at home. This preference arises from the ease of use, natural feel, realism, and intuitiveness of hand interaction, which closely resembles real-life interactions, making it particularly beneficial for patients with motor difficulties.

Ziadeh [32] explored the use of motor imagery-based BCI (MI-BCI) in stroke rehabilitation. When combined with VR, MI-BCI allows for the introduction of game-based interactions into rehabilitation. However, controlling MI-BCI can be challenging and may lead to poor performance, which can frustrate users and impact their motivation to use the technology. Users' sense of agency over the system can mitigate decreases in motivation. The study found that the avatar hand, providing a more anthropomorphic and relatable representation, significantly increased users' sense of ownership compared to abstract blocks. While both avatar hand and abstract blocks yielded similar levels of MI-BCI performance, the sense of ownership enhanced the perception of agency, which is crucial for maintaining user motivation. Designers of MI-BCI-based rehabilitation applications can leverage anthropomorphic avatars to visually represent the trained limb, thereby improving users' ownership and potentially enhancing their motivation to engage with the technology.

4) *Virtual Objects Interaction*: The size and behavior of virtual objects significantly affect individuals with multiple sclerosis. Cikajlo *et al.* [47] aimed to analyze how the size and bounce of virtual objects impact proximal and fine motor performance and intrinsic motivation in individuals with multiple sclerosis. Manipulating the size and behavior of virtual objects offers supportive tools in occupational therapy for these individuals. These interventions can improve proximal movements, fine motor skills, and reduce upper extremity disability, thus providing more engaging and challenging tasks for individuals with multiple sclerosis.

5) *Persuasive System*: The persuasive system in the study by Sanpablo *et al.* [37] involved the integration of persuasive elements into exergames for gait rehabilitation. Based on the Persuasive System Design (PSD) model, this system aimed to stimulate pediatric patients to adhere to short-term rehabilitation programs. It incorporated persuasive elements such as adjusting treadmill speed, providing auditory feedback, and introducing game challenges, all of which were adjusted

as patients progressed through their therapeutic settings. This method effectively adapted rehabilitation based on the children's performance, enhancing their motivation to engage in the rehabilitation process.

6) *Citizen Science*: In citizen science, members of the general public engage in research tasks in projects led by professional scientists. These tasks often involve data collection or analysis and do not require specific expertise [56]. Citizen science projects typically tap into intrinsic motivations, and some incorporate gaming elements like point systems or competitions to encourage long-term participation [57]. Citizen science has the potential to empower patients to contribute to scientific research, increase self-esteem, and foster a sense of community [30]. It offers intellectually stimulating and educational experiences, aligning with patient-centered rehabilitation. Citizen science projects can provide long-term motivation, especially when combined with gamified elements.

B. Social Interaction

1) *Collaboration*: Collaboration in XR-assisted rehabilitation has been shown to have a positive influence on motivation. Shah *et al.*'s study [35] emphasized the importance of collaboration in VR exergames for elderly individuals participating in rehabilitation programs. The findings demonstrated that incorporating social collaboration into VR exergames had a positive impact on participants' motivation and physical exertion levels. In comparison to individuals who played individually, participants engaged in collaborative gameplay reported significantly higher intrinsic motivation across various subscales, including enjoyment, effort, and usefulness, along with increased physical exertion. Furthermore, the participants expressed their enjoyment of the social VR gameplay and recognized the vital role that collaboration played in motivating them. This study underscores the potential of collaborative VR exergames to enhance social connectedness and engagement in physical activity among senior participants. These findings offer valuable insights and key design principles that can assist healthcare researchers, professionals, and industrial experts in developing sustainable and user-friendly VR-based exergames for elderly adults.

By prioritizing collaboration in the co-design process, the development of immersive and engaging exergaming solutions can effectively contribute to the overall well-being and rehabilitation outcomes of elderly individuals. The integration of collaboration and social interaction within SocialBike by Arlati *et al.* [36] held the potential to reduce social isolation and promote engagement in exercise and rehabilitation for frail elderly individuals. By incorporating social elements, the application enhanced users' motivation and willingness to practice. The multiplayer modes provided flexibility for users to train according to their preferences and personal attitudes. This innovative approach holds promise for addressing the challenge of low adherence to falls prevention programs in the home setting. The collaboration and social aspects of SocialBike offer opportunities for improving the overall well-being and clinical outcomes of frail older adults, as they actively participate in their rehabilitation while combating social isolation.

2) *Competition*: Competition, challenge, and working with peers have been proposed to increase intrinsic motivation. The work of Navarro *et al.* [50] found that competition might enhance the effectiveness and enjoyment of rehabilitation interventions designed to address attention deficits in individuals post-stroke, regardless of their level of competitiveness and without having a negative effect on their perceived pressure and competence.

C. Feedback

1) *Visual-Audio*: The combination of real-time verbal and visual feedback offered by MR technology holds significant potential for everyday use in physical therapy practices. This potential is based on various findings that underscore the importance of motivation in achieving long-term engagement and rehabilitation success [14]. Lancere *et al.* [14] introduced a rehabilitation technology that integrates non-invasive wireless sensors and MR to enhance muscle engagement in lower extremity amputee rehabilitation programs, specifically during deep core exercises. The study's results demonstrated that a combined approach involving verbal and visual MR feedback (VF + MR) led to a more substantial change in muscle electrical activation compared to using verbal feedback (VF) alone. These findings underscore the potential of the proposed solution to significantly increase motivation, muscle engagement, and self-awareness during the rehabilitation process, as indicated by electromyography (EMG) data and self-reported questionnaire responses.

2) *Haptic*: The use of haptic-feedback VR systems can play a significant role in rehabilitation. In their pilot study, Dong *et al.* [12] focused on the development and evaluation of a haptic-feedback VR system known as the VBBT, aiming to enhance the Box and Block Test (BBT) for more targeted assessments of upper extremity motor function. The findings highlighted that the VBBT not only provided reliable and validated evaluations of upper extremity motor function but also significantly boosted motivation during the assessment process. Specifically, it effectively identified deficiencies in hand dexterity, smoothness, and motion efficiency. The Intrinsic Motivation Inventory results revealed that patients exhibited higher enjoyment and put forth greater effort while receiving haptic feedback.

3) *Biofeedback*: Skalski *et al.* [49] employed hemodynamic (HEG) biofeedback with VR to treat attention deficits in children with attention deficit/hyperactivity disorder. HEG biofeedback provides individuals with feedback regarding cerebral perfusion (blood circulation) in the prefrontal cortex (PFC) using graphs and games, with the goal of teaching them how to consciously regulate these processes. The HEG biofeedback sessions involved controlling distractors and creating an environment that captured participants' attention. The biofeedback method employed the principles of learning and instrumental conditioning, enabling individuals to consciously regulate their cerebral perfusion. The study compared the effects of standard 2D HEG biofeedback in the lab with VR HEG biofeedback featuring limited or complex visual scenes. The VR biofeedback groups demonstrated better

performance in cognitive tests and a greater regional cerebral blood oxygenation slope during biofeedback compared to the 2D biofeedback group.

D. Rehabilitation Modalities & Participants

A significant distinction among the reviewed studies pertains to the rehabilitation modalities under investigation. Ten of the studies featured participation by healthy individuals, while the remaining studies centered on patients undergoing rehabilitation. Studies involving healthy individuals typically sought to assess the potential of XR technology in enhancing physical exercises, improving fitness, honing skill acquisition, or promoting overall well-being. Motivational factors in these studies often revolved around intrinsic rewards, enjoyment, and the pursuit of personal goals.

Conversely, studies involving patients undergoing rehabilitation faced distinct challenges. Patients in the process of rehabilitation frequently contend with health issues, pain, and the need to regain lost abilities or functions. In this context, motivation is profoundly influenced by the desire for recovery, effective pain management, and therapeutic adherence. The motivation strategies employed across these studies exhibited considerable variability. Some studies embraced patient-centered approaches, customizing XR experiences to individual patient needs, while others employed more generalized motivation techniques that could be applied to diverse rehabilitation types.

E. Intrinsic and Extrinsic motivation

Intrinsic motivation pertains to the inherent desire or internal drive that compels individuals to engage in an activity because it is inherently rewarding or enjoyable [58]. In the context of XR-assisted rehabilitation, three factors contribute to intrinsic motivation:

- **Autonomy:** Empowering users to have a sense of autonomy in their rehabilitation process by allowing them to customize exercises, set goals, and make decisions about their XR environment can promote a feeling of ownership over their rehabilitation journey.
- **Competence:** Providing users with a sense of competence or mastery over their exercises through XR experiences is crucial. Feedback mechanisms, progression tracking, and adaptive difficulty levels can help users perceive their growth and improvement.
- **Relatedness:** Incorporating social elements within XR-assisted rehabilitation can foster a sense of relatedness. Features such as multiplayer XR experiences, virtual support groups, or sharing progress with peers can enhance the feeling of connectedness and motivation.

Conversely, extrinsic motivation involves engaging in an activity for external rewards or incentives [58]. While intrinsic motivation is often preferred, extrinsic motivation strategies can also play a significant role in XR-assisted rehabilitation:

- **Rewards and Incentives:** Offering rewards or incentives for achieving rehabilitation goals can boost extrinsic motivation. These rewards can range from virtual badges

and scores to real-world incentives, such as discounts on healthcare services or fitness equipment.

- **Gamification:** Gamifying XR-assisted rehabilitation exercises by incorporating game elements, such as points, levels, and challenges, can enhance extrinsic motivation. Gamification can make rehabilitation more engaging and enjoyable, encouraging users to continue their therapy.
- **Feedback and Progress Tracking:** Providing clear feedback on users' progress can serve as an extrinsic motivator. Visualizing improvements and setting clear objectives can help users stay motivated in their rehabilitation program.

Balancing intrinsic and extrinsic motivation in XR-assisted rehabilitation presents a significant challenge. While external rewards can initially pique interest, the ultimate goal is to gradually shift users toward finding their intrinsic motivation. Designing XR experiences that commence with external incentives and then foster internal motivation could be a promising approach.

F. IVR vs NIVR

The high degree of realism achievable in IVR can render rehabilitation exercises more engaging and enjoyable, potentially amplifying user motivation to adhere to their rehabilitation regimen. Moreover, IVR permits patients to practice motions directly related to their specific condition, thus rendering therapy more pertinent. However, it's worth noting that some users may experience discomfort, cybersickness, or other side effects when immersed in VR. These issues have the potential to adversely affect motivation and should be diligently addressed.

Conversely, NIVR is often more accessible, as it eliminates the need for users to wear headsets or encounter the potential discomfort associated with IVR. This accessibility is especially critical for patients with specific physical limitations. Nonetheless, it's important to acknowledge that the reduced level of immersion in NIVR might impact the sense of presence and motivation. Consequently, the choice between immersive and non-immersive VR in XR-assisted rehabilitation should be meticulously considered based on the specific needs and preferences of the target users. The selection should align with the goals of the rehabilitation program and the comfort of the users.

G. AR vs VR vs MR

AR enhances the relevance of rehabilitation exercises to real-world tasks, VR provides immersive and controlled environments, and MR combines elements of both to create dynamic and engaging experiences. The choice of XR technology in rehabilitation should align with the specific goals of the rehabilitation program and the preferences of the users.

H. Age

Age is a significant factor to consider when striving to improve users' motivation in XR-assisted rehabilitation. It is important to recognize that individuals of different ages may have varying needs, preferences, and responses. Younger

users are more intrinsically motivated, driven by novelty and gaming elements present in XR systems [45]. In contrast, older adults exhibit different motivational patterns, often valuing rehabilitation outcomes, health benefits, and social aspects of the experience [35]. To enhance motivation in XR-assisted rehabilitation, it is essential to tailor interventions based on age-related characteristics. For younger users, incorporating gamification elements, challenges, and rewards may be effective in maintaining motivation. Older users, on the other hand, may benefit from user-friendly interfaces, social interaction features, and clear health-related feedback. Younger individuals may be more focused on returning to work or sports, while older adults may prioritize activities of daily living and independence. Understanding these goals and aligning XR-assisted rehabilitation accordingly is essential.

V. LIMITATIONS, STRENGTHS AND FUTURE DIRECTIONS OF XR-ASSISTED REHABILITATION

A. Limitations

This review incorporated a total of 28 studies, several of which exhibited relatively small sample sizes, as evidenced in studies by [14], [17], [28], [35], [36], [43], [45]. The presence of limited sample sizes in certain studies may potentially undermine the reliability of their findings. Moreover, it is essential to highlight that 10 out of the 28 studies focused on normal users rather than patients, which may affect the generalizability and applicability of the proposed solutions in the context of rehabilitation. To enhance the robustness of the conclusions drawn, it is imperative for future research to encompass larger-scale studies involving more diverse populations.

Furthermore, the current body of literature on XR-assisted rehabilitation and user motivation remains relatively limited. This scarcity of studies, especially those that meet rigorous inclusion criteria, restricted our capacity to conduct a more comprehensive and in-depth analysis. As the field progresses, forthcoming research endeavors may unveil additional strategies that were not encompassed within the scope of this review. In addition, it is important to acknowledge that most of the studies included in this review exhibited relatively short intervention periods. Consequently, the ability to evaluate the long-term effectiveness of the identified motivation-enhancing strategies remains hampered. Future research should aim to delve into the sustainability of these strategies over extended rehabilitation periods.

Moreover, it is worth noting that XR technologies are in a continual state of advancement, rendering the studies encompassed in this review potentially outdated. Therefore, future research must remain cognizant of these technological advancements and adapt motivation-enhancing strategies accordingly. The success of motivation-enhancing strategies in XR-assisted rehabilitation may hinge on various contextual factors, including the specific patient population, the severity of the condition under treatment, and the expertise of healthcare professionals involved. Unfortunately, these factors were not consistently addressed in the included studies, thus limiting the capacity to provide recommendations concerning the universality of these strategies.

B. Strengths

This review has effectively identified and synthesized key findings related to gamification, social interaction, and feedback mechanisms in the context of XR-assisted rehabilitation. The practical implications of these findings extend to researchers, healthcare professionals, and developers, offering valuable insights into designing engaging and motivating XR interventions. Through highlighting the positive effects of these identified techniques on user motivation, engagement, and rehabilitation outcomes, this review provides a clear roadmap for the development of more effective XR interventions. Ultimately, these interventions have the potential to enhance rehabilitation experiences for individuals with physical or cognitive impairments.

C. Future Directions

As previously discussed, the enhancement of user motivation through gamification entails elements such as adjusting game difficulty and enriching narrative elements. Given the ongoing advancements in AI, there exists a promising avenue for in-depth research in exploring how AI-driven adjustments to game difficulty and narrative enhancement can positively influence user motivation within the context of XR-assisted rehabilitation. Furthermore, while earlier studies predominantly emphasized human-to-human collaboration as a means to enhance motivation, there is a burgeoning need to investigate whether collaboration between individuals and AI systems might offer even more substantial advantages concerning user motivation. The precise nature of the collaboration dynamics that will emerge when humans and AI cooperate remains an intriguing area that necessitates further exploration [59].

Delving into the integration of BCI also represents a pioneering research direction within XR-assisted rehabilitation. According to Peksa *et al.* [60], BCI offer the potential to monitor real-time neural activity, enabling personalized adaptations of rehabilitation activities based on users' cognitive and emotional states. This technology can provide real-time neurofeedback, allowing the XR environment to respond to users' mental conditions, thus amplifying motivation. In future research focusing on personalized BCI, it is imperative to explore ways to tailor the feedback from the brain to each user. This entails considerations regarding which aspects of brain signals to display, how to present them, and how to ensure that the feedback remains engaging and motivating for the user. The ultimate goal is to avoid mundane feedback and, instead, furnish feedback that sustains user engagement, motivation, and satisfaction.

Furthermore, it is worthwhile to investigate studies that motivate users to perform exercises with XR assistance in non-rehabilitation settings. For instance, the incorporation of visual-audio feedback in combination with full-body motion recognition provides personalized feedback for each body part [61]. Additionally, integrating various game mechanics, including variations in game types, has shown a significant impact on critical outcomes such as focus, fitness, health improvement, and independence [62]. Exploring the implementation of these techniques within rehabilitation settings

could yield promising and unexpected results in enhancing the users motivation.

VI. CONCLUSION

This systematic review has diligently explored the techniques for enhancing users' motivation in XR-assisted rehabilitation, offering a comprehensive summary of the available evidence in this burgeoning research area. The collective findings consistently underscore the potential of XR interventions to augment users' motivation, self-awareness, engagement, and adherence to rehabilitation programs. Motivation-enhancement techniques encompass a multifaceted array of approaches. Gamification strategies encompass elements such as the manipulation of difficulty levels, narrative depth, avatar embodiment, interaction with virtual objects, persuasive systems, and the fusion of citizen science. Social interaction unfolds through the dimensions of collaboration and competition. Concurrently, the feedback mechanism embraces diverse modalities, including visual-audio feedback, haptic feedback, and biofeedback.

Nevertheless, it is imperative to acknowledge the prevailing limitations within the existing body of evidence. The constraint of small sample sizes and a focus on specific populations inevitably curtails the generalizability of the findings. The imperative need for larger-scale studies encompassing diverse populations becomes evident to fortify the level of evidence in this domain. By embarking on an ongoing journey to explore and refine these techniques, XR-assisted rehabilitation stands poised to potentially catalyze a transformation within the realm of rehabilitation, ultimately enhancing the quality of life for individuals grappling with physical or cognitive impairments.

REFERENCES

- [1] C. Andrews, M. K. Southworth, J. N. Silva, and J. R. Silva, "Extended reality in medical practice," *Current treatment options in cardiovascular medicine*, vol. 21, pp. 1–12, 2019.
- [2] A. S. Pillai and P. S. Mathew, "Impact of virtual reality in healthcare: a review," *Virtual and augmented reality in mental health treatment*, pp. 17–31, 2019.
- [3] R. S. Calabrò, A. Cerasa, I. Ciancarelli, L. Pignolo, P. Tonin, M. Iosa, and G. Morone, "The arrival of the metaverse in neurorehabilitation: Fact, fake or vision?" *Biomedicine*, vol. 10, no. 10, p. 2602, 2022.
- [4] I. M. Punt, S. Armand, J.-L. Ziltener, and L. Allet, "Effect of wii fit™ exercise therapy on gait parameters in ankle sprain patients: A randomized controlled trial," *Gait & posture*, vol. 58, pp. 52–58, 2017.
- [5] C. DiCesare, A. Kiefer, S. Bonnette, and G. Myer, "Realistic soccer-specific virtual environment exposes high-risk lower extremity biomechanics," *J Sport Rehabil*, vol. 29, no. 3, pp. 294–300, 2019.
- [6] J. C. Reneker, W. C. Pannell, R. M. Babl, Y. Zhang, S. T. Lirette, F. Adah, and M. R. Reneker, "Virtual immersive sensorimotor training (vist) in collegiate soccer athletes: a quasi-experimental study," *Heliyon*, vol. 6, no. 7, 2020.
- [7] S. Gianola, E. Stucovitz, G. Castellini, M. Mascali, F. Vanni, I. Tramacere, G. Banfi, and D. Tornese, "Effects of early virtual reality-based rehabilitation in patients with total knee arthroplasty: a randomized controlled trial," *Medicine*, vol. 99, no. 7, 2020.
- [8] A. Gokeler, M. Bisschop, G. D. Myer, A. Benjaminse, P. U. Dijkstra, H. G. van Keeken, J. J. van Raay, J. G. Burgerhof, and E. Otten, "Immersive virtual reality improves movement patterns in patients after acl reconstruction: implications for enhanced criteria-based return-to-sport rehabilitation," *Knee Surgery, Sports Traumatology, Arthroscopy*, vol. 24, pp. 2280–2286, 2016.
- [9] J. Schuermans, A. Van Hootegem, M. Van den Bossche, M. Van Gendt, E. Witvrouw, and E. Wezenbeek, "Extended reality in musculoskeletal rehabilitation and injury prevention-a systematic review," *Physical therapy in sport*, vol. 55, pp. 229–240, 2022.
- [10] K. Chatterjee, A. Buchanan, K. Cottrell, S. Hughes, T. W. Day, and N. W. John, "Immersive virtual reality for the cognitive rehabilitation of stroke survivors," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 30, pp. 719–728, 2022.
- [11] C. Gatto, G. D'Errico, F. Nuccetelli, V. De Luca, G. I. Paladini, and L. T. De Paolis, "Xr-based mindfulness and art therapy: facing the psychological impact of covid-19 emergency," in *Augmented Reality, Virtual Reality, and Computer Graphics: 7th International Conference, AVR 2020, Lecce, Italy, September 7–10, 2020, Proceedings, Part II 7*. Springer, 2020, pp. 147–155.
- [12] Y. Dong, X. Liu, M. Tang, H. Huo, D. Chen, Z. Wu, R. An, and Y. Fan, "A haptic-feedback virtual reality system to improve the box and block test (bbt) for upper extremity motor function assessment," *Virtual Reality*, vol. 27, no. 2, pp. 1199–1219, 2023.
- [13] M. Solcà, V. Krishna, N. Young, M. Deogaonkar, B. Herbelin, P. Orepic, R. Mange, G. Rognini, A. Serino, A. Rezai *et al.*, "Enhancing analgesic spinal cord stimulation for chronic pain with personalized immersive virtual reality," *Pain*, vol. 162, no. 6, pp. 1641–1649, 2021.
- [14] L. Lancere, M. Jürgen, and H. Gapeyeva, "Mixed reality and sensor real-time feedback to increase muscle engagement during deep core exercising," *Virtual Reality*, pp. 1–15, 2023.
- [15] S. M. Glegg, L. Holsti, S. Stanton, S. Hanna, D. Velikonja, B. Ansley, D. Sartor, and C. Brum, "Evaluating change in virtual reality adoption for brain injury rehabilitation following knowledge translation," *Disability and Rehabilitation: Assistive Technology*, vol. 12, no. 3, pp. 217–226, 2017.
- [16] P. Pons, S. Navas-Medrano, and J. L. Soler-Dominguez, "Extended reality for mental health: Current trends and future challenges," *Frontiers in Computer Science*, vol. 4, p. 1034307, 2022.
- [17] Y. Izountar, S. Benbelkacem, S. Otmane, A. Khababa, M. Masmoudi, and N. Zenati, "Vr-peer: A personalized exer-game platform based on emotion recognition," *Electronics*, vol. 11, no. 3, p. 455, 2022.
- [18] M. Meccawy, "Creating an immersive xr learning experience: A roadmap for educators," *Electronics*, vol. 11, no. 21, p. 3547, 2022.
- [19] J. Ratcliffe, F. Soave, N. Bryan-Kinns, L. Tokarchuk, and I. Farkhatdinov, "Extended reality (xr) remote research: a survey of drawbacks and opportunities," in *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, 2021, pp. 1–13.
- [20] G. Baltaci, G. Harput, B. Haksever, B. Ulusoy, and H. Ozer, "Comparison between nintendo wii fit and conventional rehabilitation on functional performance outcomes after hamstring anterior cruciate ligament reconstruction: prospective, randomized, controlled, double-blind clinical trial," *Knee surgery, sports traumatology, arthroscopy*, vol. 21, pp. 880–887, 2013.
- [21] H. M. Rao, R. Khanna, D. J. Zielinski, Y. Lu, J. M. Clements, N. D. Potter, M. A. Sommer, R. Kopper, and L. G. Appelbaum, "Sensorimotor learning during a marksmanship task in immersive virtual reality," *Frontiers in psychology*, vol. 9, p. 58, 2018.
- [22] M. Villiger, J. Liviero, L. Awai, R. Stoop, P. Pyk, R. Clijnsen, A. Curt, K. Eng, and M. Bolliger, "Home-based virtual reality-augmented training improves lower limb muscle strength, balance, and functional mobility following chronic incomplete spinal cord injury," *Frontiers in neurology*, vol. 8, p. 635, 2017.
- [23] W. G. Wright, "Using virtual reality to augment perception, enhance sensorimotor adaptation, and change our minds," *Frontiers in systems neuroscience*, vol. 8, p. 56, 2014.
- [24] E. Wiskerke, J. Kool, R. Hilfiker, K. M. Sattelmayer, and G. Verheyden, "Determining the optimal virtual reality exergame approach for balance therapy in persons with neurological disorders using a rasch analysis: Longitudinal observational study," *JMIR serious games*, vol. 10, no. 1, p. e30366, 2022.
- [25] D. M. Nekar, H. Y. Kang, and J. H. Yu, "Improvements of physical activity performance and motivation in adult men through augmented reality approach: a randomized controlled trial," *Journal of Environmental and Public Health*, vol. 2022, 2022.
- [26] D. Moher, A. Liberati, J. Tetzlaff, D. G. Altman, and t. PRISMA Group*, "Preferred reporting items for systematic reviews and meta-analyses: the prisma statement," *Annals of internal medicine*, vol. 151, no. 4, pp. 264–269, 2009.
- [27] A. M. Methley, S. Campbell, C. Chew-Graham, R. McNally, and S. Cheraghi-Sohi, "Pico, picos and spider: a comparison study of specificity and sensitivity in three search tools for qualitative systematic reviews," *BMC health services research*, vol. 14, no. 1, pp. 1–10, 2014.
- [28] A. Khaleghi, Z. Aghaei, and F. Hosseinnia, "Toward using effective elements in adults' amblyopia treatment in a virtual reality-based gamified binocular application," *Entertainment Computing*, vol. 43, p. 100504, 2022.

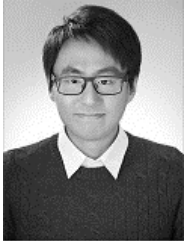
- [29] M.-C. Juan, J. Elexpuru, P. Dias, B. S. Santos, and P. Amorim, "Immersive virtual reality for upper limb rehabilitation: comparing hand and controller interaction," *Virtual Reality*, pp. 1–15, 2022.
- [30] R. Barak Ventura, K. Stewart Hughes, O. Nov, P. Raghavan, M. Ruiz Marín, and M. Porfiri, "Data-driven classification of human movements in virtual reality-based serious games: Preclinical rehabilitation study in citizen science," *JMIR Serious Games*, vol. 10, no. 1, p. e27597, 2022.
- [31] T. Chen, J. Chen, C. K. Or, and F. P. Lo, "Path analysis of the roles of age, self-efficacy, and tam constructs in the acceptance of performing upper limb exercises through immersive virtual reality games," *International Journal of Industrial Ergonomics*, vol. 91, p. 103360, 2022.
- [32] H. Ziadeh, D. Gulyas, L. D. Nielsen, S. Lehmann, T. B. Nielsen, T. K. K. Kjeldsen, B. I. Hougaard, M. Jochumsen, and H. Knoche, "'mine works better': Examining the influence of embodiment in virtual reality on the sense of agency during a binary motor imagery task with a brain-computer interface," *Frontiers in psychology*, vol. 12, p. 806424, 2021.
- [33] S. E. J. Knobel, B. C. Kaufmann, S. M. Gerber, P. Urwyler, D. Cazzoli, R. M. Müri, T. Nef, and T. Nyffeler, "Development of a search task using immersive virtual reality: proof-of-concept study," *JMIR Serious Games*, vol. 9, no. 3, p. e29182, 2021.
- [34] J. C. Rodrigues, P. Menezes, and M. T. Restivo, "An augmented reality interface to control a collaborative robot in rehab: A preliminary usability evaluation," *Frontiers in Digital Health*, vol. 5, p. 1078511, 2023.
- [35] S. H. H. Shah, A. S. T. Karlsen, M. Solberg, and I. A. Hameed, "A social vr-based collaborative exergame for rehabilitation: codesign, development and user study," *Virtual Reality*, pp. 1–18, 2022.
- [36] S. Arlati, V. Colombo, D. Spoladore, L. Greci, E. Pedrolì, S. Serino, P. Cipresso, K. Goulene, M. Stramba-Badiale, G. Riva *et al.*, "A social virtual reality-based application for the physical and cognitive training of the elderly at home," *Sensors*, vol. 19, no. 2, p. 261, 2019.
- [37] A. I. P. Sanpablo, J.-A. Armenta-García, A. F. Muñiz, A. M. Peñaloza, A. Mendoza-Arguiles, and M. D. Rodríguez, "Integration of persuasive elements into exergames: Application in the development of a novel gait rehabilitation system for children with musculoskeletal conditions," *Journal of Biomedical Informatics*, vol. 132, p. 104130, 2022.
- [38] B. Stammli, K. Flammer, T. Schuster, M. Lambert, and H.-O. Karnath, "Negami: An augmented reality app for the treatment of spatial neglect after stroke," *JMIR Serious Games*, vol. 11, p. e40651, 2023.
- [39] M. Eckert, B. Domingo Soria, and N. Terroso Gil, "Finding effective adjustment levels for upper limb exergames: Focus group study with children with physical disabilities," *JMIR Serious Games*, vol. 11, p. e36110, 2023.
- [40] R. Herne, M. F. Shiratuddin, S. Rai, D. Blacker, and H. Laga, "Improving engagement of stroke survivors using desktop virtual reality-based serious games for upper limb rehabilitation: A multiple case study," *IEEE Access*, vol. 10, pp. 46 354–46 371, 2022.
- [41] D. R. Allegue, S. N. Sweet, J. Higgins, P. S. Archambault, F. Michaud, W. C. Miller, M. Tousignant, and D. Kairy, "Lessons learned from clinicians and stroke survivors about using telerehabilitation combined with exergames: Multiple case study," *JMIR Rehabilitation and Assistive Technologies*, vol. 9, no. 3, p. e31305, 2022.
- [42] M. E. Moan, E. K. Vonstad, X. Su, B. Vereijken, M. Solbjør, and N. Skjæret-Maroni, "Experiences of stroke survivors and clinicians with a fully immersive virtual reality treadmill exergame for stroke rehabilitation: a qualitative pilot study," *Frontiers in Aging Neuroscience*, vol. 13, p. 735251, 2021.
- [43] I. Lehmann, G. Baer, and C. Schuster-Amft, "Experience of an upper limb training program with a non-immersive virtual reality system in patients after stroke: a qualitative study," *Physiotherapy*, vol. 107, pp. 317–326, 2020.
- [44] G. G. Fluet, Q. Qiu, J. Patel, A. Cronic, A. S. Merians, and S. V. Adamovich, "Autonomous use of the home virtual rehabilitation system: a feasibility and pilot study," *Games for health journal*, vol. 8, no. 6, pp. 432–438, 2019.
- [45] A. Elor, M. Powell, E. Mahmoodi, M. Teodorescu, and S. Kurniawan, "Gaming beyond the novelty effect of immersive virtual reality for physical rehabilitation," *IEEE Transactions on Games*, vol. 14, no. 1, pp. 107–115, 2021.
- [46] M. Eckert, A. Aglio, M.-L. Martín-Ruiz, and V. Osma-Ruiz, "A new architecture for customizable exergames: User evaluation for different neuromuscular disorders," in *Healthcare*, vol. 10, no. 10. MDPI, 2022, p. 2115.
- [47] I. Cikajlo, A. Hukić, and A. Udovčić Pertot, "The size and behavior of virtual objects have influence on functional exercise and motivation of persons with multiple sclerosis: a randomized study," *Scientific Reports*, vol. 12, no. 1, p. 19375, 2022.
- [48] O. Stamm, S. Vorwerg, M. Haink, K. Hildebrand, and I. Buchem, "Usability and acceptance of exergames using different types of training among older hypertensive patients in a simulated mixed reality," *Applied Sciences*, vol. 12, no. 22, p. 11424, 2022.
- [49] S. Skalski, K. Konaszewski, G. Pochwatko, R. Balas, and J. Surzykiewicz, "Effects of hemoencephalographic biofeedback with virtual reality on selected aspects of attention in children with adhd," *International Journal of Psychophysiology*, vol. 170, pp. 59–66, 2021.
- [50] M. D. Navarro, R. Llorens, A. Borrego, M. Alcañiz, E. Noé, and J. Ferri, "Competition enhances the effectiveness and motivation of attention rehabilitation after stroke: a randomized controlled trial," *Frontiers in human neuroscience*, vol. 14, p. 385, 2020.
- [51] A. R. Jafari, A. A. Shafiee, A. Mirzajani, and P. Jamali, "Cam visual stimulation with conventional method of occlusion treatment in amblyopia: a randomized clinical trial," *Tehran University Medical Journal*, vol. 72, no. 1, 2014.
- [52] G. Rasch, *Probabilistic models for some intelligence and attainment tests*. ERIC, 1993.
- [53] C. D. Batson, P. A. Van Lange, N. Ahmad, D. A. Lishner, M. Hogg, and J. Cooper, "Altruism and helping behavior," in *The Sage handbook of social psychology*. Sage London, UK, 2007, pp. 241–258.
- [54] M. O. Wilhelm and R. Bekkers, "Helping behavior, dispositional empathic concern, and the principle of care," *Social psychology quarterly*, vol. 73, no. 1, pp. 11–32, 2010.
- [55] E. L. Deci and R. M. Ryan, "Self-determination theory," *Handbook of theories of social psychology*, vol. 1, no. 20, pp. 416–436, 2012.
- [56] O. Nov, O. Arazy, and D. Anderson, "Scientists@ home: what drives the quantity and quality of online citizen science participation?" *PLoS one*, vol. 9, no. 4, p. e90375, 2014.
- [57] C. T. Callaghan, J. J. Rowley, W. K. Cornwell, A. G. Poore, and R. E. Major, "Improving big citizen science data: Moving beyond haphazard sampling," *PLoS biology*, vol. 17, no. 6, p. e3000357, 2019.
- [58] C. P. Silverthorne, *Organizational psychology in cross cultural perspective*. NYU Press, 2005.
- [59] M. Muller, L. B. Chilton, A. Kantosalo, C. P. Martin, and G. Walsh, "Genaichi: generative ai and hci," in *CHI conference on human factors in computing systems extended abstracts*, 2022, pp. 1–7.
- [60] J. Peksa and D. Mamchur, "State-of-the-art on brain-computer interface technology," *Sensors*, vol. 23, no. 13, p. 6001, 2023.
- [61] P. Caserman, S. Liu, and S. Göbel, "Full-body motion recognition in immersive-virtual-reality-based exergame," *IEEE Transactions on Games*, vol. 14, no. 2, pp. 243–252, 2021.
- [62] A. W. De Vries, J. H. Van Dieën, V. Van Den Abeele, and S. M. Verschueren, "Understanding motivations and player experiences of older adults in virtual reality training," *Games for health journal*, vol. 7, no. 6, pp. 369–376, 2018.



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